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Ports, Routes and Boats

*A Three-Stage Optimization Framework for Urban Waterborne Parcel
Logistics in Bergen*

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Abstract

As urban parcel demand rises, trucks face growing limits in dense city centres. However, in Bergen, the same geography that constrains road delivery also creates another delivery channel: water. This thesis examines how a waterway parcel network could be designed for Bergen, with ports acting as parcel-locker locations. The network is designed through a three-stage optimization framework. First, a partial set covering model selects port locations. Second, a symmetric capacitated boat routing model builds delivery routes under capacity and shift limits. Third, a scheduling model assigns the routes to boats and crews. Using household data, parcel-demand data, OpenStreetMap data, and manually measured maritime-distance data, the thesis compares two port layouts and tests low, average, and peak demand days across three boat capacities and three crew sizes. Arna, Fana, and Åsane are excluded after the walking-time analysis, since the candidate ports could not cover them within a realistic walking time. The results therefore refer to the remaining central and coastal districts of Bergen. The results show that 20 to 28 ports give similar walking outcomes, with average walking times around 20 min. Operations, however, are mainly driven by demand. The network requires 1 to 10 boats and 10 to 76 employees depending on the demand scenario, with peak days forcing the system to scale significantly. A 1500-parcel boat offers the best capacity compromise, while larger crews are most useful during peak periods. The thesis shows that waterborne parcel logistics in Bergen could be feasible in principle.

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1 Introduction

Global parcel volumes have increased substantially over the last decade, largely driven by e-commerce, and several industry forecasts expect continued growth toward 2030 (Pitney Bowes 2023; World Economic Forum 2020). At the same time, conventional truck-based delivery faces increasing operational constraints in many urban centres (Bosona 2020; Maxner et al. 2022). Cities increasingly regulate freight access through measures such as low-emission zones, delivery time windows, and pedestrian areas, while dense or historic street networks can also limit the suitability of large vehicles (Ahani et al. 2023; Milenković et al. 2024; Verlinde et al. 2016). The last mile is therefore becoming both more constrained and more expensive at the very moment its volume reaches historic highs.

If trucks have less room to operate while parcels still need to reach customers, the question becomes which alternative delivery channel can fill this gap. One common response is to move to smaller vehicles such as electric vans and cargo bikes, which work well in dense urban areas where deliveries are clustered, but scale poorly in cities that are larger, hillier, or more spread out (Narayanan and Antoniou 2022). Another answer predates these vehicles entirely. Some cities have moved goods on water for most of their existence, and several are now reviving the practice (DHL Group 2025; Fludis 2026).

Located on Norway’s western coast, Bergen combines a constrained urban road network with a geography shaped by fjords, making it a relevant candidate for such a system. This thesis therefore examines the implications of implementing a boat-based parcel delivery network in Bergen, in which selected shoreline ports would serve as parcel locker locations. Boats would operate along predefined routes between these ports to deliver parcels, while residents would collect their parcels by walking to their nearest assigned port.

1.1 Problem Description

Designing a waterborne parcel network for a city like Bergen is not one isolated decision, but a sequence of connected choices. Three connected questions have to be answered together for the system to make sense as a whole.

The first question is where the boats should stop. Parcels arriving by water still need to reach the customers who ordered them, and this final step is completed on foot. The ports therefore have to be placed close enough to where people live, ensuring that the walking distance required to collect a parcel remains reasonable. Opening a port in every coastal neighbourhood would solve the coverage problem but waste resources, while keeping only

a few central ports would push too much walking onto residents. The first part of the problem is to find a set of port locations that balances population coverage against the number of ports.

The second question is how the boats move between these ports. Once ports are chosen, each one carries a share of the city's parcel demand. The boats then have to leave the depot, visit the ports, drop off the parcels and return. The routes also have to respect realistic capacity and shift limits.

The third question is how many boats and crews are actually needed. A set of routes is not yet a plan. The routes have to be packed into boats and shifts so that the network can operate every day of the year, including peak days when demand spikes. The fleet size drives the capital investment of the network, the crew count drives its labour cost, and both depend on the routes coming out of the previous stage.

These three problems are linked but not the same. Solving them in isolation would either produce an unrealistic plan that ignores the city's geography, or a well-placed set of ports that no fleet can practically serve. This thesis treats them as a single chain where each stage takes the output of the previous one as its input.

1.2 Research Question

Building on the problem above, this thesis asks the following main research question:

How can a waterborne parcel delivery network be designed for the city of Bergen, and what fleet and crew would such a network need to operate?

To make this question addressable, it is broken into three sub-questions, each one matching one step of the methodology:

1. How many ports are needed, and where should they be placed, to cover the population of Bergen within a reasonable walking time?
2. Given a set of selected ports and their parcel demand, what set of boat routes can serve them while respecting realistic capacity and shift limits?
3. Given these routes, how many boats and crews are required to operate the network across low, average and peak demand days?

Together, the answers to these three sub-questions give a complete picture of what the network would look like.

1.3 Scope

This thesis focuses on the design of the network and its operational requirements rather than on a full real-world implementation. Several boundaries are drawn to keep the problem tractable.

On the demand side, port locations are defined using household locations only. As a result, the total network demand considered in the model is limited to household demand.

From an operational perspective, the depot location is selected manually based on its proximity to Bergen's existing port and rail infrastructure. Candidate port locations are also evaluated at a conceptual level only. They are considered suitable if they can potentially be served by boat, while a detailed site-level feasibility assessment is outside the scope of this thesis. Consequently, factors such as land ownership, operating rights, available quay space, water depth and handling capacity are not explicitly evaluated.

Several broader implementation considerations are also outside the scope of this thesis. Although the environmental impact of replacing trucks with boats is one of the main motivations for the topic, it is not quantified in the analysis. The financial cost of the network is also not modelled, due to the absence of reliable cost data for boat operation in this specific context. These aspects are instead highlighted as relevant directions for future work.

1.4 Thesis Structure

The remainder of the thesis is organised as follows. Section 2 reviews the relevant background, covering urban parcel delivery and the Bergen-specific context. Section 3 presents the literature review of the optimization problems used to address the research question. Section 4 describes the data used in the thesis. Section 5 explains the three optimization models that form the core of the analysis, namely the port selection model, the boat routing model and the boat scheduling model. This section also describes the construction of the demand scenarios used to test the proposed network. Section 6 then presents and discusses the results obtained from the three models, including the selected port layouts, the generated routes, the fleet and crew requirements for each demand

scenario, and the response of the main metrics to changes in boat capacity and crew size. Finally, Section 7 summarises the findings, answers the research question, and outlines directions for future work.

2 Background

This section sets the context for the rest of the thesis. It first looks at urban parcel delivery and explains why the last mile has become the most expensive and complex part of the supply chain. It then turns to Bergen, describing the geography that makes road delivery difficult and the city's ongoing push towards zero-emission mobility. Finally, it presents waterborne transport as an alternative for urban freight, along with the speed restrictions that apply on Bergen's waterways.

2.1 Urban Parcel Delivery and Last-Mile Logistics

With global parcel volumes expected to reach 225 billion in 2028, up from 161 billion in 2022 (Pitney Bowes 2023), delivery companies will have to adapt quickly as annual parcel volumes continue to grow. Over long distances, capacity can still be added through more container ships, trains, and trucks. In city centres, however, there is only so much space for trucks. The supply chain can scale in response to rising demand, but its final bottleneck is simple: as goods move closer to customers, the vehicles serving each link get smaller (Government Office for Science 2019).

At the end of the chain, on what is called the *last mile*, the job falls to medium and small trucks (Government Office for Science 2019). The last mile is widely recognised as the most expensive part of the supply chain, often accounting for more than half of the total delivery cost (Savelsbergh and Van Woensel 2016). It is also the most operationally complex. Routes have many short stops, vehicles are rarely full, and time windows are tight. Trucks also have to navigate dense city environments that are often poorly adapted to modern freight vehicles, especially in European historic centres where narrow streets, restricted access and limited road space make urban deliveries more difficult (Bachofner et al. 2022). Combined with demand spread across the whole city, this forces trucks to be smaller, while they still need to make many deliveries to reach all customers, which in turn requires more trucks.

Meanwhile, cities that are often already experiencing record congestion from private car use are now also facing more delivery trucks on their roads (Lindsey et al. 2023). Delivery vehicles contribute disproportionately to urban congestion relative to their share of total traffic (International Transport Forum 2022). They are harder to electrify than cars, so they continue to generate high levels of CO₂ and noise pollution while operating close to where people live (Coulombel et al. 2018). Even when they are electric, trucks are much heavier than passenger cars, which means more wear and tear on roads and higher

levels of dust pollution (World Bank 2022). Many of these costs are external to the carrier. The carrier pays for fuel and labour, while the city and its inhabitants pay for the rest (European Commission n.d.).

In response, some cities and governments are creating projects and regulations to reduce these impacts. The research field that groups these efforts together is known as *city logistics* (Savelsbergh and Van Woensel 2016). The goal is simple: replace one long, polluting last leg of delivery with a shorter, cleaner, and often shared one. This can be done in many ways.

One option is to set up micro-consolidation centres at the edge of dense areas, from which cargo bikes or other small electric vehicles can handle the last segment. If this is not feasible, truck traffic can be moved to off-peak hours to spread it out. Another option is to restrict high-emission vehicles from entering certain parts of the city, often the dense and walkable city centre (CIVITAS 2015).

Finally, a solution that can create value for all parties is to consolidate household deliveries into small, shared pickup points across the city. This makes shipping companies more efficient since more demand is assigned to a single delivery spot, and it allows delivery trucks to serve only a few concentrated locations, often in commercial areas. For residents, this offers the option to pick up parcels when the parcels are ready, rather than having to wait at home for delivery. In Norway, both PostNord and Posten have already rolled out automated parcel lockers along these lines, and they are now used by a large share of the population (Hovi et al. 2023; Posten Norge 2026; PostNord Norge 2026).

Many of these solutions involve the participation of both shipping companies and the population. The population plays a crucial role, because it often has to cover a portion of the delivery route itself. Even if a customer orders a package and would prefer to have it delivered to their address, the package may instead need to be picked up somewhere else, close to its original delivery point. Ideally, this pickup would not generate additional emissions. For this to work, the pickup points need to be reasonably close to where everyone lives. As more cities face the same challenge, a planning concept known as the *15-minute city* has emerged (Moreno et al. 2021). The idea is simple. All essential services should be reachable within a 15 min walk for every resident. This thesis adopts the same 15 min threshold as the base walking time for its port selection model.

2.2 Bergen as a Constrained Logistics Environment

This thesis is based on the city of Bergen, located on the west coast of Norway. The city sits directly between the North Sea and the mountains that frame it. Bergen is famously known as the city of seven mountains, and these iconic peaks, combined with the fjords that cut deep into the urban area, leave only narrow strips of land where the population can live. People therefore mostly live close to the coast, while the land rises quickly into high mountain valleys further inland. The city is split into eight districts (*bydeler*): Arna, Bergenhus, Fana, Fyllingsdalen, Laksevåg, Ytrebygda, Årstad and Åsane.

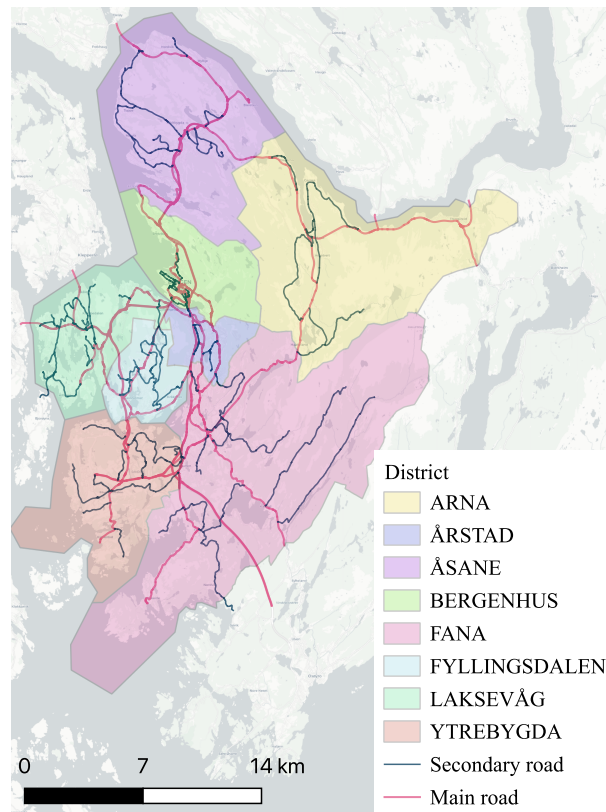


Figure 1: Map of Bergen with its districts and road network.

Figure 1 shows the geographic distribution of these districts, and the full population breakdown by district is shown in the Appendix in Figure A.18. They can be grouped into two sets. The first group, made up of Bergenhus, Årstad, Fyllingsdalen, Laksevåg and Ytrebygda, sits in the central and coastal part of the city, around the main fjords, and accounts for about 65 % of Bergen's total population. The second group, Åsane, Arna and Fana, lies on the outer edges of the municipality. Because this second group has a much larger area with lower population density and mostly mountainous terrain, its road network is also less dense than in the first group.

Bergen also has significant elevation changes across its urban area, again caused by the

surrounding mountains that force the population into limited coastal areas. Because flat coastal land runs out quickly, part of the population lives at higher elevations on the slopes leading up from the shore (ULaaDS 2022; Venvik et al. 2016). This makes walking distance in Bergen harder to evaluate, as slope strongly affects walking speed. According to Tobler’s Hiking Function (Tobler 1999), a walk that takes 15 minutes on flat ground stretches to 21 minutes at a 10 % incline. Walking is fastest at a 5 % decline, where the same walk takes only 12.5 minutes, but steeper declines of 10 % bring the pace back down to 15 minutes. Standard pedestrian network analysis tends to ignore elevation (Sevtsuk and Kalvo 2025), even though it is known to have a real effect on walking time in hilly cities like Bergen (Barthelemy et al. 2025). Since the goal of this thesis is to explore the feasibility of a waterway shipping system rather than to plan a real operational schedule, slope is ignored in the walking time calculations and identified as a direction for future work.

Moreover, Bergen is known as one of Europe’s rainiest cities (Bremer et al. 2020). Frequent rain leads to wet pavements and infrastructure conditions that could make long walking distances less attractive, as does snow in winter. Nevertheless, as with slope, this limitation is ignored in this work, under the assumption that Bergen’s residents are accustomed to local weather conditions. In a real-world scenario, it would be useful to compare the impact of weather on walking to port locations with its impact on driving.

The same geography that constrains road delivery offers a natural alternative. Because the population lives mostly along the coast, residents are already close to potential waterways. Road routes, on the other hand, must work around Bergen’s geography. They either go around mountains and fjords, cross them with bridges, or cut through them with tunnels. Every workaround adds distance and creates a chokepoint. For boats, those same fjords are direct, open corridors between districts. What slows down road delivery is exactly what makes water delivery efficient.

Bergen also has the existing infrastructure to support such a system. As the second largest city in Norway, it hosts both the second largest port (Port of Bergen 2026) and the second largest cargo rail station (Bane NOR 2026) in the country, both located in the central area of the city. In practice, this means that the heavy logistics needed to feed a waterborne parcel system are already in place and would not need to be built from scratch.

2.3 Bergen as a Testbed for Zero-Emission Urban Mobility

Beyond its geography and existing infrastructure, Bergen is also actively changing how transport works inside the city. The city has set the goal of an emission-free city centre by 2030, and rather than banning cars outright, it focuses on shared and electric mobility (Invest in Bergen 2019). As part of this effort, Bergen became the first city in Norway to open dedicated mobility hubs, which combine reserved spaces for shared electric cars with public transport access, bicycle parking and charging points. Since a single shared car is estimated to replace up to ten privately owned ones (Invest in Bergen 2019), the city sees this as a way to free up a large amount of road space while still keeping motorised transport available to everyone.

Moreover, in March 2025, the Norwegian government announced that it wanted to allow cities to introduce zero-emission zones, following requests from Oslo, Bergen and Trondheim (Taylor 2025). The rules still had to be prepared and sent for consultation, but the direction is clear: diesel freight vehicles may face tighter access to central Bergen, which strengthens the case for cleaner ways of moving parcels.

Bergen is also being used directly to test these alternatives. For example, through the Shift2Zero project, the electric vehicle maker Paxster is working with DHL and the City of Bergen to test zero-emission last-mile delivery in shared pedestrian and vehicle zones (Paxster 2025). The pilot pairs vehicles that automatically adjust their speed depending on location with a modular “swap box” system that removes the need for large micro-depots (Paxster 2025). Bergen is therefore already testing clean urban logistics in practice, and a road-free delivery channel such as the waterborne network studied in this thesis would fit naturally into this direction.

2.4 Waterborne Transport as an Urban Freight Alternative

While the trucking industry today dominates urban freight to the point where it can seem impossible to imagine cities working without it, large urban centres had to organise their cargo flows for centuries before trucks existed. For example, the Netherlands and England relied on their canal networks to move goods into and out of their cities. Before mechanised land transport, waterways were central to freight movement, but this began to change with the rise of canals and railways during the Industrial Revolution (Rodrigue 2024). In recent years, several companies have started to revive this more traditional way of handling cargo. DHL is running waterway delivery trials in cities like Venice, Lagos and Berlin (DHL Group 2025), while Fludis is operating on the Seine in Paris (Fludis 2026),

both with the goal of reducing the number of trucks in city centres and the pollution that comes with them. The Paris case illustrates the approach well. The city is dense and difficult for trucks to navigate, which gave Fludis the opportunity to use the Seine as a backbone for its operations. The idea is simple. Electric delivery bikes are first loaded onto a barge-like boat. During the boat's journey, parcels are transferred to the bikes, and the boat carries both to an upriver drop-off point. From there, the bikes handle the last-mile delivery and then return to the boat for the trip back to the main warehouse. This setup removes the short-range constraint of the bikes while keeping trucks out of the city itself.

Bergen has several characteristics that make this approach especially relevant. Where Paris works with a single river, Bergen has a network of fjords and inlets that already touches almost every populated district. Using these waterways to reach the population by boat would free the streets from the polluting trucks that currently handle this work. A boat network would not be subject to peak-hour traffic, to the road bottlenecks created by Bergen's terrain, or to the geographic constraints that make some parts of the city hard to reach by road in the first place. On top of this, the fjords cut deep into the city in several places, which allows boats to reach areas that sit well inland from the open coast.

2.5 Bergen Waterway Speed Restrictions

Bergen has two official maritime speed-restriction zones, shown in Figure 2. These zones were obtained directly from The Norwegian Coastal Administration (Kystverket 2026). The pink zone corresponds to a speed limit of 5 knots, while the green zone corresponds to a speed limit of 8 knots. The port-to-port distances calculated later in the thesis account for both restricted-speed zones.

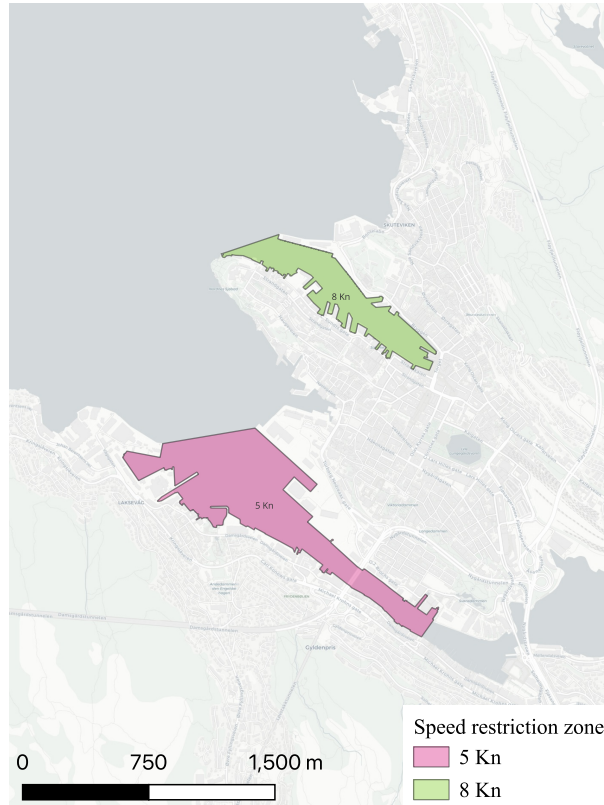


Figure 2: Bergen speed restriction zones.

No additional speed restrictions were included in the model. Norwegian maritime regulations require vessels to maintain a safe speed and to navigate without creating danger to life, health, the environment, or material values (Lovdata 1975, 2007). However, these requirements are context-dependent rather than fixed spatial speed zones. Therefore, only the official mapped 5-knot and 8-knot restriction zones obtained from Kystinfo were included in the port-to-port travel time calculations.

3 Theory

This section reviews the optimization problems behind the three models used in this thesis. It first covers facility location problems, which underpin the port selection model. It then presents the vehicle routing problem, used to build the boat routes, and finally the vehicle scheduling problem, used to assign those routes to boats and crews.

3.1 Facility Location Problems

The Facility Location Problem (FLP) family covers optimization models that decide where to open facilities so that a set of demand points is served in the best possible way. The classical setup takes a set of candidate locations, a set of demand points, and a distance or cost for every demand-facility pair, and then asks which subset of locations should be opened. What counts as “best” depends on the objective, which usually balances the cost of opening facilities against the cost of serving demand from them.

The literature has converged on a small number of classical formulations. The p -median problem, introduced by Hakimi in 1964, opens exactly p facilities and minimises the total weighted distance between demand points and their nearest facility. The p -center problem, also introduced by Hakimi in 1965, has the same structure but minimises the maximum distance instead of the total, which makes it useful when worst-case service matters. The Set Covering Location Problem (SCLP) of Toregas et al. (1971) takes a different angle. It fixes a coverage distance, considers a demand point covered if a facility is open within that distance, and minimises the number of open facilities under the constraint that every demand point is covered. The Maximum Covering Location Problem (MCLP) of Church and ReVelle (1974) relaxes the requirement to cover every demand point, fixes the number of facilities at p , and maximises the demand covered within the distance threshold.

The Partial Set Covering Location Problem (PSCLP) sits between the SCLP and the MCLP. Like the SCLP it minimises the number of open facilities and uses a coverage distance to define what counts as served. Like the MCLP it accepts that not every demand point will be covered. The constraint is instead that a target share, or absolute amount, of demand must be covered (Cordeau et al. 2019). The PSCLP is therefore well suited to settings where full coverage is either too expensive or simply not realistic, and where the planner can specify a service level rather than a strict requirement.

Covering models are common in last-mile logistics, where deciding where to put pickup points, parcel lockers, or micro-depots is a central part of network design. Schwerdfeger

and Boysen (2020) use covering ideas to place mobile parcel lockers in urban areas, while Deutsch and Golany (2017) study the location of pickup points under capacity constraints. The trade-off between the number of facilities and the share of customers within a comfortable walking distance is a recurring theme, which makes the partial covering family a natural fit (Boyer et al. 2009).

3.2 Vehicle Routing Problem

The Vehicle Routing Problem (VRP) family builds on the classic Travelling Salesman Problem (TSP). The TSP consists of finding the minimum-distance route that visits each customer once and returns to its starting point (Dantzig, Fulkerson, et al. 1954).

The VRP was first formulated by Dantzig and Ramser (1959) as a generalisation of the TSP to the case where a fleet of vehicles, instead of a single one, must serve a set of customers from a central depot. Since then the VRP has grown into one of the most studied problems in combinatorial optimization, with a large literature on both modelling and solution techniques (Vidal et al. 2020).

Many variants of the VRP have appeared over the years, each one adding features as it was used in new applications. For example, the Capacitated Vehicle Routing Problem (CVRP) gives each vehicle a maximum load and forces the total demand on a route to stay below that limit. The Vehicle Routing Problem with Time Windows (VRPTW) requires every customer to be visited within a given interval. The multi-depot VRP allows several depots, the pickup and delivery VRP combines both directions of flow, and the heterogeneous fleet VRP works with vehicles of different types and capacities. A second distinction that matters for this thesis is between symmetric and asymmetric travel cost matrices. VRP models may be formulated on either directed or undirected transport networks (Braekers et al. 2016). In this thesis, the waterway network is treated as symmetric, meaning that the travel time between two ports is assumed to be the same in both directions.

Moreover, if the constraint requiring each customer to be visited only once is too restrictive for the context, two approaches can be considered. The first approach is to relax this constraint directly. This leads to the Split Delivery Vehicle Routing Problem (SDVRP), where vehicles keep the same capacity constraints as in the CVRP, but customers may be visited more than once when their demand cannot be served by a single vehicle or route (Archetti et al. 2011). The second approach is to keep the original CVRP formulation and modify the input data instead, often called the *a priori split strategy* (Torkzaban et al. 2024). In this case, a maximum demand limit is defined, and customers with demands

above this limit are divided into several artificial customer nodes. Each clone represents part of the original customer demand and respects the imposed demand limit. These cloned customers keep the same distance values to all other customers, while the distance between clones of the same original customer is set to zero. This provides a practical simplification that allows a standard CVRP model to approximate split deliveries without requiring a full SDVRP implementation. More specific splitting strategies can be used to further optimise the model, as uniformly splitting the demand could result in suboptimal solutions (Torkzaban et al. 2024).

For this thesis, because the SDVRP can be more computationally expensive (Torkzaban et al. 2024), many different configurations are tested, and the number of ports and customer demands is relatively small, the chosen model is the Symmetric Capacitated Vehicle Routing Problem (SCVRP) with the *a priori split* strategy.

3.3 Vehicle Scheduling Problem

The Vehicle Scheduling Problem (VSP) usually appears after the routing decisions in transport planning. In the standard planning hierarchy used in public transport and urban freight, decisions move from strategic choices about the network, to tactical choices about which routes to operate and how often, and finally to operational choices about which vehicle does which trip and when (Ibarra-Rojas et al. 2015). Vehicle scheduling sits at the boundary between tactical and operational planning. Once the set of trips to run is known, the VSP turns those trips into a usable schedule by matching them with the available fleet.

When the trips are fixed loops from depot to depot and the only question is how to assign these loops to vehicles within a fixed duration, the VSP can be formulated as a bin packing problem (Delorme et al. 2016). Each vehicle is represented as a bin with a capacity equal to the available working duration, and each loop is represented as an item with a size equal to its duration. The objective is to use the minimum number of vehicles needed to operate all loops.

In the context of this thesis, the number of routes to assign on a given day is expected to be small, while the number of available boats is not fixed in advance. The objective is therefore to determine the minimum number of boats required to operate all planned loops within the maximum working duration of each boat. For the main case study, a complete enumeration of all possible route-to-boat assignments would probably be sufficient. However, this approach does not scale well as the number of routes increases.

Therefore, the problem is implemented as a bin packing optimization model, so that the same formulation can also be used in potential cases where a larger set of daily routes must be scheduled, for which a range of metaheuristic methods is also available (Munien and Ezugwu 2021).

4 Data and Operational Constraints

The data used in this thesis come from three independent sources, each addressing a distinct aspect of the problem: the spatial distribution of Bergen’s population, the parcel demand it generates, and the pedestrian network connecting households to candidate ports. Population data are provided at the household level by Bergen municipality for the year 2020. Parcel demand is based on PostNord’s weekly volumes by postal code in 2020. Finally, the walking network of Bergen is extracted from OpenStreetMap (OSM). Additionally, this section outlines specific operational constraints. The remainder of this chapter describes these sources in detail and the steps taken to prepare them for use in the models that follow.

4.1 Walking Network of Bergen

OSM is first used to retrieve the walking graph of Bergen with the Python package `OSMnx` (Boeing 2025) and its `graph_from_place()` function. This package can retrieve the graph of a specific place (`place_name='Bergen, Norway'`) and network type (`network_type='walk'`). However, during spot checks of walking distances, the network type filter was found to be too narrow, as Google Maps walking distances were generally shorter than the distances calculated from the graph. After investigation, the network filter was found to exclude all bicycle lanes, even when they were labelled as walkable by OSM. The filter was therefore changed to allow bicycle lanes when they are walkable.

4.2 Population and Household Data

The household data were obtained from the Master’s thesis by Fang and Opedal (2020) and originally came from Bergen municipality. The 2020 dataset contains information about the city’s population at the household level. Only the coordinates, number of residents, postal code, and district were kept from the original data. The dataset contained 149 308 households (73 393 unique), with a total population of around 280 491. This population is split across Bergen’s eight districts and 142 unique postal codes.

To reduce the size of the data, the household coordinates were snapped to the nearest OSM node in the walking network. This reduced the dataset to 29 991 unique OSM nodes. During this aggregation, some OSM nodes were assigned populations from multiple districts, representing around 418 people (0.15 % of the population) and 27 nodes (0.07 %

of nodes). Because some results are shown by district, these nodes were excluded. The population distribution by district can be seen in Figure 3. It should be noted that some districts are filtered out of the population data in Section 5.1.1 (Walking Thresholds).

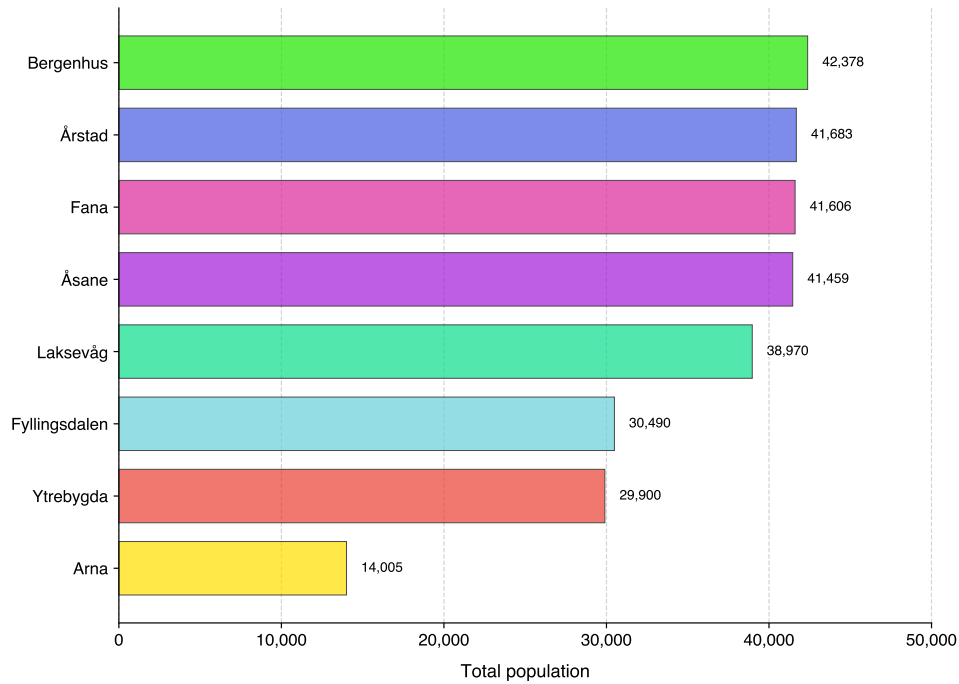


Figure 3: Total population of Bergen in 2020 aggregated by district.

Since the goal of this dataset is to identify good port candidates based on population coverage rather than absolute population size, it is not updated to current figures. Only the spatial distribution is taken into account, and it is assumed that population growth has been geographically uniform.

4.3 Candidate Ports

To retrieve potential port locations, `OSMnx` was again used. The `features_from_place()` function can filter potential features by OSM tags. Table A.11 shows the filter tags used to retrieve port candidates based on the OSM documentation (OpenStreetMap contributors 2026).

The resulting data contain 746 port locations. The distribution of the ports by the tags used in Table A.11 can be found in Table A.12 of the Appendix. As previously done with the household coordinates, the port locations were snapped to the nearest OSM node on the walking graph. This reduced the data to 390 candidate port nodes. These locations were then projected on a map and manually filtered to remove locations that were not suitable for this thesis. For example, some nodes were in a body of water that was not

Table 1: Candidate and excluded ports by district.

District	Candidate Ports	Excluded Ports
Ytrebygda	130	24
Bergenhush	47	9
Laksevåg	41	6
Fana	31	23
Fyllingsdalen	21	6
Åsane	18	10
Årstad	17	0
Arna	5	2

accessible from the main waterways, or were extremely close to another node that was more suitable. A total of 80 nodes were removed, for a final total of 310 port nodes. The kept and removed nodes are shown in Figure 4. The breakdown of candidate and excluded ports by district is shown in Table 1.

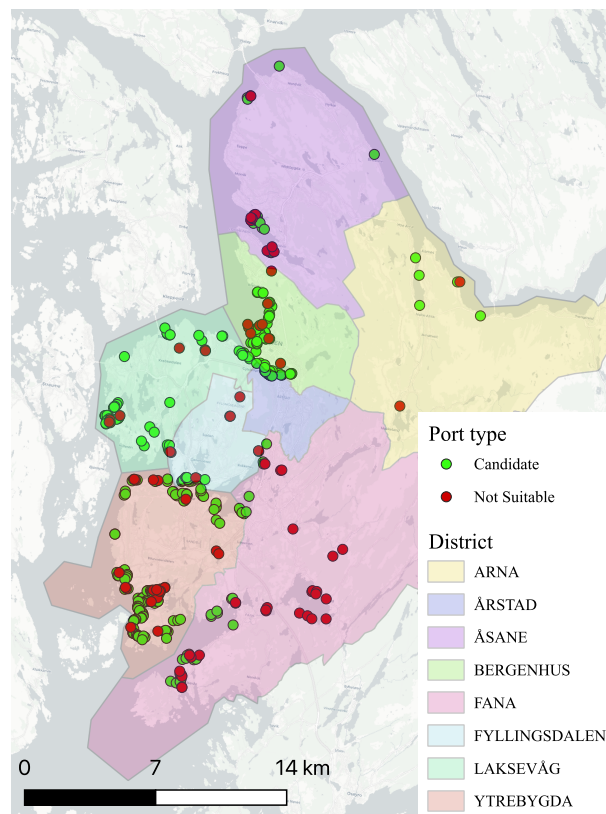


Figure 4: Map of the port nodes across Bergen's eight districts. Nodes kept for the analysis are shown as candidates, while the nodes removed by hand are marked as not suitable.

4.4 Distance and Travel Time Data

Having defined the household-node and port-node datasets, the distance matrix can now be built. To achieve this, the Python package `networkx` (Hagberg et al. 2008) was used to calculate the shortest distance between all combinations of port and household nodes using its `single_source_dijkstra_path_length()` function. This resulted in an $m \times n$ matrix D , with $d_{i,j}$ representing the Dijkstra shortest-path distance between the i -th port and the j -th household node¹. For this thesis, the distance between any $port_i$ and $household_j$ is assumed to be symmetric ($d_{i,j} = d_{j,i}$).

To convert distances into walking times, an average walking speed first had to be defined. Bosina and Weidmann (2017), which aggregated walking speed data from more than 200 measurements reported in the literature, gives a reference free-flow speed of 1.34 m/s. Giannoulaki and Christoforou (2024), in a systematic review of walking speed studies, reports a slightly lower value of 1.25 m/s as the speed commonly recommended for pedestrian design. Since these two reference values are close, a walking speed of 1.3 m/s, approximately their midpoint, was adopted as the general assumption in this thesis. This value is therefore consistent both with the average speed observed in empirical studies and with the value used in pedestrian design practice.

No walking speed study specific to Bergen was available, and the actual speed is likely to vary with season and weather and, in particular, with the city's topography. Giannoulaki and Christoforou (2024) further notes that walking speed decreases on inclined surfaces and is significantly affected by weather, both of which are relevant for Bergen given its hilly terrain and wet climate. Because the magnitude of these effects is difficult to quantify for the specific paths considered, the value of 1.3 m/s was retained. The resulting population coverage as a function of walking time, with all candidate port nodes available, is shown in Figure 5.

4.5 PostNord Parcel Demand Data

Parcel demand data were obtained from Orhan et al. (2024), but originated from PostNord. They cover weekly shipment and parcel volumes, from week 1 to week 48, by postal code for the year 2020. The dataset had demand for 439 unique postal codes. It contained information about the number of shipments and parcels. However, in many cases, the number of shipments was larger than the number of parcels in a row. Demand was therefore calculated as the minimum of the two columns.

¹All pairs were reachable by walking due to the OSM node snapping and filtering of the port nodes.

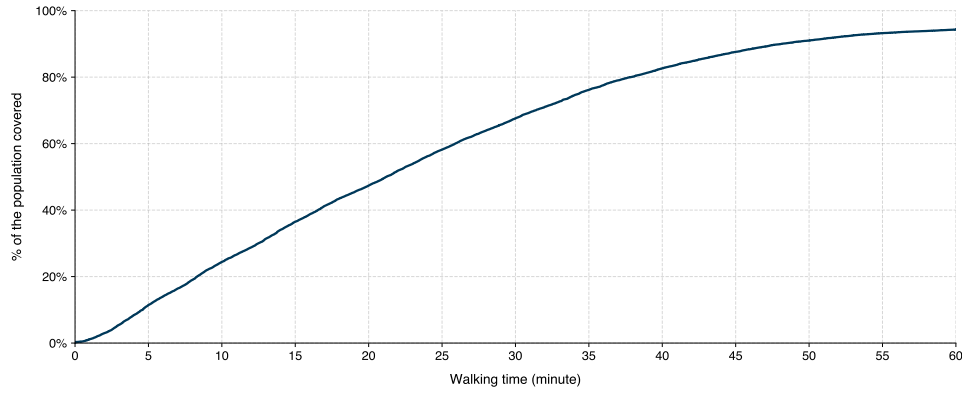


Figure 5: Percentage of the population covered by walking time with all ports available.

Unlike the household data, the real demand size is important for optimising port capacity. Because the PostNord dataset is from 2020, it needs to be adjusted to reflect 2026 levels. This was achieved using PostNord’s annual reports, which give the total amount of parcel deliveries for each reporting year. PostNord’s yearly demand can be found in Table A.14 in the Appendix. Based on these yearly demands, the 2026 Bergen demand is derived in two steps. First, the 2025 national demand is projected one year forward to 2026 using the Compound annual growth rate (CAGR). Due to the COVID-19 pandemic, only the years 2022 to 2025 are used to calculate this rate, starting from a year in which demand had begun to normalise after the pandemic (Pitney Bowes 2023). As shown in Equation A.17, the rate is 4.68 % per year, which brings the 281 million parcels of 2025 to about 294 million in 2026. Second, the 2020 Bergen demand is scaled by the ratio between this projected 2026 national demand and the reported 2020 national demand of 198 million, assuming that Bergen’s share of the national demand stays constant. This ratio represents a growth of around 48.56 % since 2020, which is applied to the 2020 Bergen figure to approximate its 2026 level.

Second, because the dataset represents only PostNord’s share of the market in 2020, it needs to be adjusted to current market figures. The *Report on the Use of Postal Services by Private Individuals and Businesses in Norway, 2025*² (Nkom 2026), published by the Norwegian Communications Authority, does not provide exact volume-based market shares for parcel deliveries, but it surveys consumer usage patterns across Norway’s primary delivery companies. PostNord’s market share was therefore derived from the normalised results of the private market parcel survey and estimated at 19.61 %. The survey data can be found in Table A.13 in the Appendix.

Third, this dataset represents parcel demand for both households and other types of customers. However, since this thesis focuses on populations, the non-household demand

²Translated from the Norwegian: *Rapport om privatpersoners og bedrifters bruk av posttjenester i Norge - 2025*.

needs to be removed. To achieve this, the household postcode data were joined with the weekly demand, and postcodes not present in the household dataset were excluded.

Finally, since some districts are filtered out of the population data in Section 5.1.1 (Walking Thresholds), the demand associated with these excluded household districts is also removed. The overall demand quantities and changes by step can be seen in Table 2. The total parcel demand considered in this thesis is 5 977 449 parcels for the year 2026.

Table 2: Sequential adjustments and filtering of parcel demand volumes.

Data Cleaning / Scaling Step	Parcel Volume	Change
PostNord 2020 Baseline	1,628,441	–
PostNord 2026 Projection	2,419,134	+790, 693
With Total Market Share (19.61%)	12,336,226	+9, 917, 092
Only Households	8,125,871	–4, 210, 355
Operational Dataset (Filtered Districts)	5,977,449	–2, 148, 422

4.6 Disaggregating Weekly Demand into Daily Demand

Since daily scheduling for the boat network requires daily demand for each selected port, the PostNord weekly demand needs to be broken down. Several approaches were possible, including equal daily splits or random allocation. This thesis applies weekday-specific rates because they better reflect real demand patterns. Based on the work of Lorenzo-Espejo et al. (2024), which studied daily parcel operations, the 5-day split for this thesis is obtained by adding the weekend demand to Monday. The resulting daily rates can be seen in Table A.15 in the Appendix.

4.7 Disaggregating Postal Code Demand into Household Demand

To correctly assign demand to the nearest chosen ports, the demand of each household must be known. However, the PostNord dataset is only available by postal code. Nevertheless, demand can still be allocated to each household based on its share of the population within its postal code. For example, if a postal code has a weekly demand of 100 parcels and contains two households, one with 1 resident (a) and the other with 2 (b), the demand is distributed as $\frac{1}{3}$ to a and $\frac{2}{3}$ to b . It should be noted that this is a simplification for this thesis and is only used as an assumption. In addition, each household demand was

rounded to an integer, resulting in a loss of 15.69 total parcels or 0.000 263 % of total demand.

4.8 Depot Location

With the candidate ports and their demand defined, the depot location can now be chosen. The depot is where incoming parcels are received, processed, and loaded onto the boats, so it needs enough space for this work and for docking the boats. However, these processing and handling spaces are considered only as context and are not quantified in this thesis. Furthermore, a good depot location should be close to the Port of Bergen and its cargo rail station, as both are the second largest of their kind in Norway (Bane NOR 2026; Port of Bergen 2026). Finally, the depot also acts as a port itself, so a location with a high nearby population is the most efficient choice, as it removes the most parcels from boat transport.

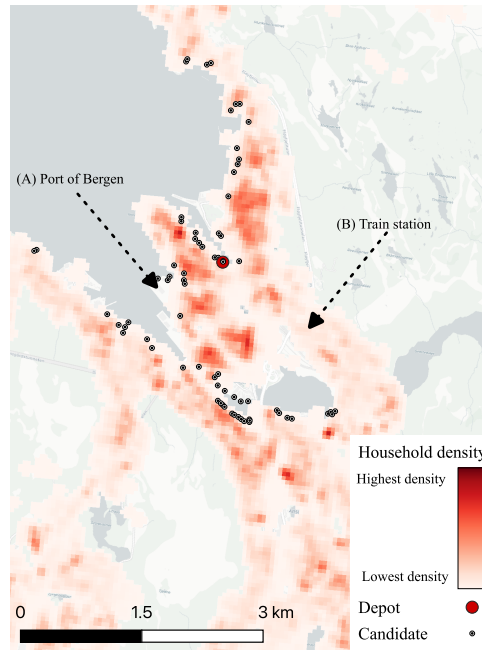


Figure 6: Map of Bergen city centre showing household density in 2020 and the locations of candidate ports relative to the (A) Port of Bergen and the (B) train station.

The first step consists of identifying the locations of the Port of Bergen and the railway station, and overlaying them with household density. As shown in Figure 6, the two facilities are spatially separated. The port is located along the shoreline, whereas the railway station is situated further inland. Since the depot also needs to be close to the shore, the range of potential locations becomes considerably restricted. A full density map can be found in Figure A.19 of the Appendix.

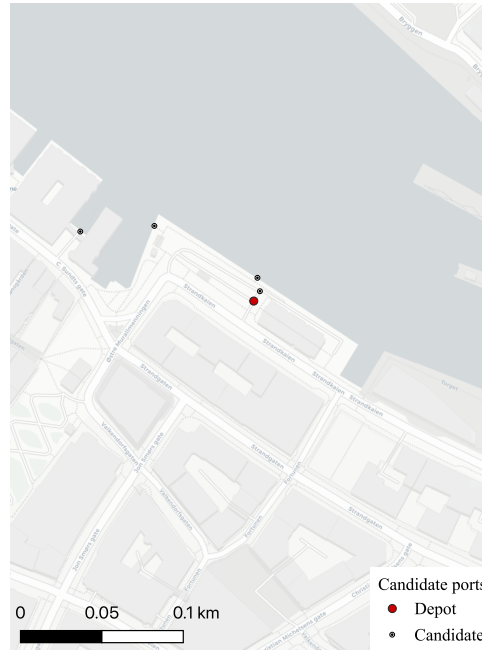


Figure 7: Map of the selected depot location.

The chosen location is shown in Figure 7. It is directly in a high-density area, close to the port, and relatively near the rail station. Additionally, it offers some empty space near the shore. As explained earlier, this space may not actually be usable for this project, but as for the candidate ports, it is assumed that all locations offer enough space and could be built on.

4.9 Operational Constraints

The models presented later in this thesis are based on a set of constraints that must be defined and justified before they can be implemented. The following section details each constraint and explains the values assigned to them. Some operational constraints, such as handling time at the depot or travel time between ports and lockers, are excluded from the scope of this thesis. Including these constraints would require a more detailed specification of the operational process and more precise port-to-locker distance data. Despite these exclusions, the selected constraints can still provide a reasonable foundation for the development of the models.

4.9.1 Boat Maximum Capacity and Speed

This thesis uses three scenarios to simulate different types of boats in the network, each representing a different total parcel capacity, from low to high. An average parcel weight

of 4.5 kg is used as a modelling assumption. This value lies between the 4 kg average parcel weight reported by Stålhammar et al. (1996) and the 5 kg or lower average reported for most parcel operators in the Irish parcel delivery sector by Frontier Economics (2021).

The low-capacity scenario is based on the Candela P-12 Shuttle boat (Candela 2026). Although this boat is not designed for cargo transport, it is used as a representative vessel class to provide realistic characteristics for the models. Its service speed of 25 knots, 30-passenger capacity, and total payload limit of 3000 kg are taken directly from the manufacturer's specifications (Candela 2026). To determine the parcel capacity, the passenger weight limit is used as a baseline. Assuming a standard passenger weight of 75 kg (Norwegian Maritime Authority 2026), the payload dedicated strictly to parcels is 2250 kg. With an average parcel weight of 4.5 kg, this yields a total capacity of 500 parcels. The remaining 750 kg of the boat's total 3000 kg limit is then allocated to the handling materials required for these parcels. Dividing this remaining capacity by the 500 parcels yields a handling material weight of exactly 1.5 kg per parcel. This 1.5 kg per parcel ratio is used across the other scenarios to maintain a consistent proportion of handling weight to loading capacity.

The high-capacity scenario is based on the Kewatec Work 1920 (Oy Kewatec AluBoat Ab 2023), a larger vessel made to transport cargo of many types. The total load capacity can be taken directly as 15 000 kg, with a cruising speed of 25 to 35 knots. With this capacity, the same average parcel weight of 4.5 kg per parcel, and 1.5 kg per parcel for handling, the total parcel capacity is exactly 2500 parcels.

The mid-capacity scenario is based on the Kewatec Work 1500 (Oy Kewatec AluBoat Ab 2026), a vessel whose physical size sits directly between the small and large boat choices. Because specific payload data are not published by the manufacturer, a capacity of 1500 parcels is assumed to create a clean middle ground between the other two scenarios. As for the other capacity scenarios, the boat's speed is specified by the manufacturer and is also 25 to 35 knots.

For all scenarios, boat range is not considered in this thesis, since the routes are not fully optimised waterway navigation routes. It is therefore assumed that boats could stop to refuel or use strategically placed electric chargers at high-demand ports and the depot while unloading.

4.9.2 Crew Size

Each boat is operated by a crew. The crew is responsible for discharging the parcels at the ports. Crew size is tested with 2, 3, and 4 employees per boat. A minimum crew of two persons per boat is assumed, since single-handed operation is generally not recommended for small commercial workboats unless the vessel is specifically approved for it (Maritime and Coastguard Agency 2023). The maximum of four is used to create three practical scenarios.

4.9.3 Manoeuvring Time

Every time the boat travels from one port to another, it needs time to dock and undock. This thesis accounts for it with a fixed manoeuvring time, added to the travel time of every trip between two ports. The value used is a baseline of 3.5 min, taken as the average of two estimates: a conservative one of about 5 min, derived from a study of port operations on a passenger ferry (Kløvning 2024), and an optimistic one of 2 min, based on the Candela P-12 (Candela 2026).

4.9.4 Discharge Time

The discharge time is the time needed to handle one parcel once the boat has docked. This thesis uses a fixed value of 25 s per parcel (Hovi et al. 2023). The total discharge time at a port is then the number of parcels delivered there, multiplied by 25 s and divided by the crew size, since the employees on board discharge the parcels in parallel. Together with the travel and manoeuvring times, this discharge time makes up the total duration of a route.

4.9.5 Operating Hours

To schedule the operations correctly, the operating hours must first be defined. Since this thesis focuses on Bergen, Norway, information from parcel service providers operating in the region, such as PostNord and Posten, can be used as a reference. Both providers offer parcel delivery services on weekdays, from Monday to Friday, with some exceptions made by Posten for certain types of parcels on Saturdays (Posten Norge 2026; PostNord Norge 2026). Therefore, the operating days in this thesis are also Monday to Friday.

In addition, the daily operating hours must be specified. In this case, the Norwegian Working Environment Act provides a useful reference point. According to the Act, “Ordinary working hours must not exceed nine hours in a 24-hour period” (Stortinget 2005). However, since this represents the maximum legal limit, a standard maximum of 8 h per employee is assumed in this thesis. Therefore, any given boat route, from the moment it leaves the depot to the moment it returns, cannot last longer than 8 h. This upper bound represents a normal working shift for the crew. The overall operation is nevertheless modelled over a 24 h planning horizon, allowing several employees or shifts to cover the full day if required.

5 Methodology

In this section, the methodology of this thesis is explained in detail, in three separate parts. First, the model used to select a set of ports based on a desired walking distance is presented. Second, the model used to generate a set of routes for these ports is described. Finally, the section explains how a schedule can be constructed, given a set of ports and routes.

5.1 Port Selection Model

The goal of this stage is to find the smallest set of ports that still covers a target share of the population within an acceptable walking time. The selection depends on two inputs: the walking threshold and the target coverage percentage. These two configurations are presented first, followed by the optimization model chosen to perform the selection.

5.1.1 Walking Thresholds

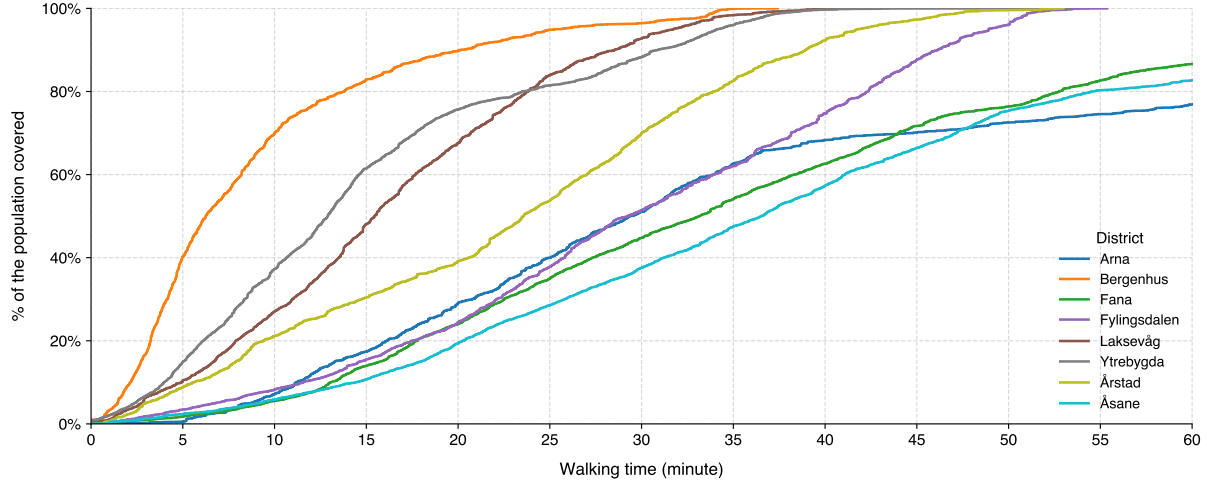


Figure 8: Percentage of the population covered by walking time with all ports available by district.

As discussed in the background section, a walking time of 15 min is often a target when planning new city programmes for residents. The walking time coverage distribution was already shown in Figure 5, but it is also important to show the same data by district. This can be visualised in Figure 8.

The districts can be visually grouped into three categories based on how their population coverage evolves with the walking time threshold. The first group, composed of

Bergenhus (■), Ytrebygda (■), and Laksevåg (■), shows a rapid increase in coverage, reaching near-full coverage within approximately 35 to 40 min of walking. The second group, composed of Årstad (■) and Fyllingsdalen (■), follows an intermediate pattern: their coverage initially grows more slowly than in the first group, but eventually reaches 100 % at higher walking thresholds, around 50 to 55 min. The third group, formed by Arna (■), Fana (■), and Åsane (■), shows the opposite trend: coverage grows slowly and does not reach 90 % even at a one-hour walking threshold, plateauing between 75 to 85 %.

Table 3: Walking time statistics in minutes by district, assuming all ports are open.

District	Avg. Time	Std. Dev.	Median Time
Arna	38.76	28.02	29.38
Åsane	38.07	19.29	36.52
Fana	37.13	24.26	33.18
Fyllingsdalen	29.37	12.87	29.23
Årstad	22.74	12.19	23.56
Laksevåg	16.13	8.67	15.44
Ytrebygda	14.83	9.78	12.90
Bergenhus	8.91	7.60	6.26

The same pattern appears in Table 3: Arna, Fana, and Åsane have the highest average walking times to the nearest port by a clear margin, all around 37 to 39 min, while every other district stays below 30 min. These three also show the largest standard deviations, which means some residents do live close to a port, but many others live much further out than the average suggests, so access across these districts stays poor overall. Based on these observations, and combined with the points discussed in the background section, Arna, Fana, and Åsane are excluded from the thesis, as the port locations cannot reach them within a realistic walking time.

The walking time threshold considered for this problem ranges from 15 to 30 min, with every integer value within this range tested. Values below 15 min were judged as too low for this context, as the resulting coverage was very limited, while values above 30 min were judged as impractical, since citizens are unlikely to walk such distances to collect a parcel.

5.1.2 Coverage Percentages

Since achieving 100 % coverage for a given walking time threshold could require a disproportionately large number of ports, due to certain households only being reachable through specific ports, a minimum coverage percentage is defined instead. The range is set to 75 to 100 %, with every percentage point within this range tested. This range should be large enough to show if some ports are added only for a small portion of the population.

5.1.3 Partial Set Covering Location Problem

Although various models exist to address this covering problem, the PSCLP was selected for this thesis. It matches the network's operational objectives as it minimises the number of active ports while relaxing the requirement to reach all households (Cordeau et al. 2019).

Table 4 presents the model configurations, all combinations of which were tested.

Table 4: Configurations for the PSCLP.

Symbol	Description	Value
α	Target demand coverage threshold	$\{0.75, 0.76, \dots, 1\}$
$T_{\max}$	Maximum allowable walking time limit (minutes)	$\{15, 16, \dots, 30\}$

Most of the model implementation comes directly from the Cordeau et al. (2019) paper, although some changes were made to fit this thesis. Table 5 details the notation used in the model. The changes made in the notation were mostly done to contextualise it for the thesis, as household and port nodes are used instead of demand nodes and facilities, while population is used instead of weight. The *Data and Sets* section of the table describes what the model needs as input, with the set of port and household nodes and the population for each household. The reachable household nodes J are calculated using the previously defined distance matrix M , where every household node without a close enough port node is removed from consideration. The maximum distance used is derived from the walking speed and the current run $T_{\max}$, resulting in a maximum number of meters allowed to reach a possible port.

Building upon the established notation, the problem is formulated as an optimization model as follows:

$$\min \sum_{i \in I} f_i y_i \quad (1)$$

$$\text{s.t.} \quad \sum_{i \in I(j)} y_i \geq z_j \quad j \in J \quad (2)$$

$$\sum_{j \in J} d_j z_j \geq D \quad (3)$$

$$y_i = 1 \quad i \in P \quad (4)$$

$$y_i \in \{0, 1\} \quad i \in I \quad (5)$$

$$z_j \in \{0, 1\} \quad j \in J \quad (6)$$

Table 5: Notation for the PSCLP.

<i>Data and Sets</i>		
Symbol	Description	
I	Set of candidate port nodes, indexed by i	
J	Set of reachable household nodes, indexed by j	
d_j	Household population associated with node j	
$I(j)$	Subset of ports $i \in I$ capable of reaching household node j in $T_{\max}$	
P	Subset of preselected ports that must be opened, currently the depot	
<i>Parameters</i>		
Symbol	Description	Value
D	Minimum total household population to be covered	$\alpha \sum d_j$
f_i	Fixed cost of opening port i	1
<i>Decision Variables</i>		
Symbol	Description	Value
y_i	$\begin{cases} 1, & \text{if port } i \text{ is opened} \\ 0, & \text{otherwise} \end{cases}$	Binary $\in \{0, 1\}$
z_j	$\begin{cases} 1, & \text{if demand node } j \text{ is covered} \\ 0, & \text{otherwise} \end{cases}$	Binary $\in \{0, 1\}$

The objective function (1) traditionally minimises the total fixed cost of the opened ports. Since every candidate port is given the same opening cost ($f_i = 1$), this is the same as minimising the number of ports selected for the network.

Constraints (2) ensure that for a household node j to be covered ($z_j = 1$), at least one port from its subset $I(j)$ must be opened. If no port able to reach j within the walking threshold $T_{\max}$ is opened, z_j is forced to remain uncovered ($z_j = 0$).

Constraint (3) ensures that the covered set of household populations is greater than or equal to the minimum covered population D .

Constraints (4) force every preselected port to be opened. The depot is part of this set, as the boats operate out of it and it must therefore always be among the selected ports (Section 4.8).

The last two sets of constraints, (5) and (6), keep the port and coverage variables binary.

5.1.4 Model Implementation

The PSCLP is implemented with Google OR-Tools (Perron and Furnon 2025), using its constraint-programming solver CP-SAT (Perron and Didier 2025). One adjustment was needed to fit the model to the solver. CP-SAT only works with integer coefficients, but the coverage target $D = \alpha \sum_{j \in J} d_j$ is usually not a whole number, since α is a fraction. To account for this, the population and the coverage target are multiplied by 1000 and rounded to integers before the demand constraint (3) is built. This keeps three decimal places of precision, so the rounding has no meaningful effect on the result.

The model is kept as compact as possible by removing, before the optimization, all households that cannot be covered by any candidate port within the walking threshold $T_{\max}$. These households are therefore excluded from the formulation, and no associated coverage variable z_j is introduced.

5.1.5 Scenario Selection

Solving the PSCLP for every combination of walking threshold $T_{\max}$ and coverage target α produces a large number of runs, far too many to carry through the routing and scheduling steps that follow. Two runs are therefore kept: the best run at the 15 min threshold, which matches the “15-minute city” reference from the background section, and the best run

overall.

Deciding which run is “best” is not trivial, because each run differs along three criteria that pull against each other: the share of the population covered (α), the number of ports opened, and the average walking time to the nearest port. The number of ports reflects cost, since no financial data are available to price each one, so a network with fewer ports is treated as a cheaper one. Coverage and walking time reflect the service residents receive: coverage counts how many people fall within the walking threshold, while the average walking time also captures how far everyone has to walk, including the households the threshold leaves out. A run with higher coverage usually needs more ports, so no single run is the best on all three at once. The selection therefore relies on a *utopia point*. It is an imaginary, ideal run that takes the best value of every criterion at the same time: the highest coverage, the fewest ports, and the shortest walk. Such a run almost never exists in practice, which is why it is called a utopia, but it gives a fixed target to measure against. To apply it, the three criteria are first rescaled to a common 0–1 range, and the run that is kept is the one closest to the utopia point in Euclidean distance. This rule is the Euclidean compromise solution (Voorneveld et al. 2011), and selecting runs this way is referred to as the *utopia point method* throughout the rest of the thesis. It is applied twice: once among the 15 min runs, which gives the first scenario, and once across all runs, which gives the second. Both scenarios are then carried forward to the boat routing models in the next section.

5.2 Boat Routing Models

5.2.1 Demand Allocation to Selected Ports

The PSCLP gives a set of selected ports, but it does not specify which households each port should serve. Each household is assigned to the nearest selected port based on the distance matrix M . The total household demand is then summed by port to obtain weekly demand for each port.

It should be noted that this is done for every household, not only for the ones the PSCLP counted as covered. A household that was left uncovered, meaning that no selected port can be reached within the walking threshold $T_{\max}$, is still assigned to its nearest selected port, only with a longer walk. The selected ports therefore receive the parcel demand of every household in the included districts, and the coverage threshold α only shapes which ports are selected and how far people have to walk, not the amount of demand the network has to carry. Keeping all demand was a deliberate choice. If a network like

this were implemented, the model assumes that all parcels in the included districts would be handled by the waterborne network. This is why the network is sized to serve every household in the included districts rather than only the well-covered ones, and why the fleet and crew results in Section 6 are based on this full demand.

What changes with distance is not whether a household is served, but how it reaches its port. The walking-time results, and the coverage threshold α , assume that residents walk. This is realistic for the well-covered majority, but less so for households far from any port, who would more likely reach their port by another mode such as cycling (with a cargo basket) or public transport rather than not collect their parcels at all. The long walking times in the tail of the distribution should therefore be read as a worst-case, on-foot view of the access effort, rather than the mode every resident would actually use.

5.2.2 Port Distance Matrix

Although the computation differs from the household-to-port distance matrix, the same matrix-based logic is used here. A separate port-to-port distance matrix is constructed to represent the travel distances between each pair of selected ports. However, in complex environments like the fjords in Bergen, calculating true maritime distances between close ports is difficult. Standard Euclidean distances are inadequate, as linking two ports with a straight line could cross land masses and ignore the complex, narrow waterways of the real boat route.

Although more advanced methods exist to calculate maritime distances around coastline constraints, implementing a full shortest-path model for Bergen's coastline is outside the scope of this thesis. Given that the PSCLP reduces the problem to a relatively small set of selected ports, it was decided to use a simpler manual approach instead. This is considered sufficient for the objective of the thesis, which is to estimate network-level boat travel times rather than to provide exact navigation routes or operational sailing instructions.

The construction is done in two steps. The first step is done manually in QGIS. The selected ports are first grouped into a small number of visual zones based on their geographic position. The relevant distances are then drawn directly on top of the OSM basemap as polylines that follow a plausible boat path between two ports, and the length of each polyline is taken as the distance in nautical miles. Three sets of arcs are then drawn. The first connects every pair of ports located within the same zone. The second connects each pair of zone polygons using a single edge-to-edge polyline. The third connects each port to the depot. Since the objective is to estimate network-level travel times, inter-zone movements are approximated through representative edge-to-edge connections rather

than by computing every possible port-to-port distance directly. The 5 knots and 8 knots speed-restricted segments introduced in the background section are also included at this stage. For each polyline, the portions crossing these restricted zones are identified and assigned the corresponding travel speed.

The second step is done in code and fills in the arcs that were not drawn by hand. The directly measured arcs from the first step are loaded as the edges of a weighted graph over the selected ports. For every port pair that has no direct measurement, the shortest path through the graph is computed with Dijkstra's algorithm and used as an estimate of the missing distance. An overhead correction of 0.35 nmi³ is applied to each intermediate stop on the path to account for the fact that a boat going directly between two ports would not have to detour through every port on the way. The output is a complete port-to-port distance matrix, combining the directly measured arcs from the first step and the graph-based estimates from the second one, which is the matrix consumed by the routing models.

5.2.3 Demand Scenarios

The routing models do not run on the raw daily port demand, but on three representative daily demand scenarios per port: a Low, an Average and a Peak scenario. Working with discrete scenarios rather than the full daily series keeps the number of routing runs manageable while still covering a quiet, a typical and a busy day, which is enough to test whether the fleet sized for an average day can absorb seasonal swings.

The three scenarios are anchored on three reference months. For every month of the year, the median of total daily demand is calculated, and the months are ranked by this median. The month with the lowest median is taken as the *low month*, the one in the middle of the ranking as the *average month*, and the one with the highest median as the *peak month*. Using months instead of individual days smooths out single-day noise, like holidays or weather events, while keeping the seasonal signal.

Once the three reference months are fixed, each scenario is built at the port level. The Low scenario takes, for each port, the median of its daily demand observed during the low month, which corresponds to a typical quiet day. The Average scenario does the same on the average month and represents a typical day on which the network should comfortably operate. The Peak scenario takes the 95th percentile of the port's daily demand observed

³This value was estimated from a small set of manual checks, by drawing polylines between selected ports and the nearest existing route. It is therefore used as an approximate correction factor rather than an exact navigational distance.

during the peak month. The 95th percentile is preferred to the maximum to represent a realistically busy day without sizing the fleet on a single extreme outlier. Each scenario value is rounded up to the next integer so that the routing models treat demand as whole parcels.

5.2.4 Model Implementation

The first routing model used is the SCVRP, as it is a natural fit for this problem: the depot is fixed, each selected port has a known daily parcel demand, and the fleet is composed of identical boats with a maximum parcel capacity and a maximum daily working time. The objective is to design a set of routes, each starting and ending at the depot, that together serve all port demands while minimising the total travel time of the fleet.

Several configurations were tested for the SCVRP, each combining a boat capacity, a crew size and a daily demand scenario, summarised in Table 6. The three boat scenarios and their parcel capacities are defined in Section 4.9.1, the three crew sizes in Section 4.9.2, and the three demand scenarios in Section 5.2.3. The boat cruising speed is fixed across all configurations at 25 knots, which is the common achievable speed of every vessel considered in Section 4.9.1.

Table 6: Configurations tested for the SCVRP.

Symbol	Description	Values
Q	Boat parcel capacity	{Low-capacity, Mid-capacity, High-capacity}
e	Employees per boat	{2, 3, 4}
s	Daily demand scenario	{Low, Average, Peak}

The classic SCVRP mathematical model is not defined in this thesis, as it was also not explicitly defined in the code. The OR-Tools routing library (Perron and Furnon 2025) handles much of the standard formulation internally, such as the routing decision variables, flow conservation, and sub-tour elimination. What follows first describes the objective and the constraints layered on top of the classic model, and then the solver and search parameters used to solve it. Table 7 summarises the additional notation used in the rest of this section.

The objective minimises the total travel time of the fleet, where the cost of every arc is the sum of the boat travel time t_{ij} between the two ports and the manoeuvring time m . Parcel handling time is left out of the optimization function because it is constant, as all demand is handled at the same speed regardless of the route. To allow split deliveries while keeping the standard SCVRP structure, where each stop is visited by exactly one vehicle,

Table 7: Notation used in the SCVRP model.

Symbol	Description	Value
t_{ij}	Boat travel time between ports i and j	From port distance matrix
m	Manoeuvring time per arc	3.5 min (See Section 4.9.3)
τ	Discharge time per parcel	25 s (See Section 4.9.4)
d_i	Parcel demand at port i	From demand matrix
$T_{\max}^r$	Maximum route duration	8 h (See Section 4.9.5)

the demand of every port is decomposed into multiple service nodes that share the port's coordinates, each carrying at most 10 parcels⁴. The routing model treats these service nodes as independent stops but with no travel cost between them, making it possible for the same port to be visited several times by different routes.

The handling time at port i is $d_i\tau/e$, with d_i the parcel demand of the port for the run, τ the per-parcel discharge time from Section 4.9.4, and e the crew size of the configuration. It represents an estimate of the time needed to put the parcels into the lockers at the port. As mentioned above, the travel time between the port and the locker is not taken into account in this thesis.

Two capacity constraints are added to the solver. The first corresponds to the maximum parcel capacity of the boat in the current configuration, denoted by Q . This constraint ensures that the total demand assigned to a route does not exceed the boat capacity. The total demand is calculated by summing the demand covered by the boat at each port visited in the current route.

The second constraint corresponds to the maximum route duration, denoted by $T_{\max}^r$. This constraint represents an operational restriction requiring employees to complete their route before the end of their shift. The total route duration includes travel time, manoeuvring time, and handling time at each visited port.

The search is configured with `PARALLEL_CHEAPEST_INSERTION` as the first-solution strategy and `GUIDED_LOCAL_SEARCH` as the local-search metaheuristic (Google Developers 2026), with a time limit of 1 h per run. This combination produces a feasible solution quickly and then improves it iteratively; the best solution found within the time limit is retained.

⁴The threshold of 10 parcels was chosen after sensitivity testing. In the tested scenarios, increasing the maximum service-node demand beyond this value did not improve the routing solutions.

5.3 Boat Scheduling Model

The SCVRP gives the set of routes that the fleet needs to run on a given day, but not how many boats and crews are needed to run them. The boat scheduling model fills this gap by assigning each route to a boat and deciding how many crew shifts each boat has to run.

A simple bin packing model (Delorme et al. 2016) is enough at this stage. The routes from the SCVRP already start and end at the depot and respect the maximum route duration, so two routes on the same boat can be run one after the other with no scheduling conflict. The network also does not commit to specific delivery times at the ports, so no time windows have to be enforced. The model therefore only has to decide which routes share a boat and how many shifts each boat needs. The boat count is minimised first, and the total shift count second, since the boat count sets the capital cost of the network and the shift count sets the labour cost.

Each route is treated as a single fixed-duration, indivisible task equal to its total route time. Boats are interchangeable, and a single boat can run several routes during the day, as long as the total work fits within the 24 h planning horizon from Section 4.9.5. A crew, however, is limited to one shift of at most 8 h, so a boat that works more than eight hours in the day must operate over several shifts, each covered by a new crew. Because a route cannot be split across shifts, the model assigns every route to a specific shift of a specific boat, and not only to a boat. Table 8 summarises the notation used in the formulation.

The formulation adapts the assignment-based bin packing model of Delorme et al. (2016). The route-to-boat assignment and the per-boat duration limit follow its standard structure, while the shift dimension, the lexicographic boat-then-shift objective, and the symmetry-breaking constraints are added for this setting. The model is stated as follows:

Table 8: Notation for the boat scheduling model.

<i>Data and Sets</i>		
Symbol	Description	
R	Set of routes returned by the SCVRP, indexed by r	
K	Set of available boats, indexed by k , with $ K = R $	
s	Shift index within a boat, $s \in \{1, \dots, \bar{n}\}$	
δ_r	Total duration of route r	
<i>Parameters</i>		
Symbol	Description	Value
H	Operating day length	24 h
S	Maximum shift length	8 h
$\bar{n}$	Maximum number of shifts per boat ($\lceil H/S \rceil$)	3
M	Large weight on the boat count in the objective	10 000
<i>Decision Variables</i>		
Symbol	Description	Value
y_{rks}	$\begin{cases} 1, & \text{if route } r \text{ is assigned to shift } s \text{ of boat } k \\ 0, & \text{otherwise} \end{cases}$	Binary $\in \{0, 1\}$
u_k	$\begin{cases} 1, & \text{if boat } k \text{ is used} \\ 0, & \text{otherwise} \end{cases}$	Binary $\in \{0, 1\}$
z_{ks}	$\begin{cases} 1, & \text{if shift } s \text{ of boat } k \text{ is used} \\ 0, & \text{otherwise} \end{cases}$	Binary $\in \{0, 1\}$

$$\min \quad M \sum_{k \in K} u_k + \sum_{k \in K} \sum_{s=1}^{\bar{n}} z_{ks} \quad (7)$$

$$\text{s.t.} \quad \sum_{k \in K} \sum_{s=1}^{\bar{n}} y_{rks} = 1 \quad r \in R \quad (8)$$

$$\sum_{r \in R} \sum_{s=1}^{\bar{n}} \delta_r y_{rks} \leq H u_k \quad k \in K \quad (9)$$

$$\sum_{r \in R} \delta_r y_{rks} \leq S z_{ks} \quad k \in K, s \in \{1, \dots, \bar{n}\} \quad (10)$$

$$z_{ks} \leq u_k \quad k \in K, s \in \{1, \dots, \bar{n}\} \quad (11)$$

$$u_k \geq u_{k+1} \quad k \in K \setminus \{|K|\} \quad (12)$$

$$z_{ks} \geq z_{k, s+1} \quad k \in K, s \in \{1, \dots, \bar{n} - 1\} \quad (13)$$

$$y_{rks} \in \{0, 1\} \quad r \in R, k \in K, s \in \{1, \dots, \bar{n}\} \quad (14)$$

$$u_k \in \{0, 1\} \quad k \in K \quad (15)$$

$$z_{ks} \in \{0, 1\} \quad k \in K, s \in \{1, \dots, \bar{n}\} \quad (16)$$

The objective function (7) minimises the number of boats used first and the total number of shifts second. The fleet size is prioritised because each additional boat is a capital investment, while a shift only adds a crew for the day. The weight M on the boat count is set well above the maximum possible total shift count $|K| \cdot \bar{n}$, so the solver can never trade a shift saving for an additional boat.

Constraints (8) force every route to be assigned to exactly one (boat, shift) pair.

Constraints (9) limit the total workload of boat k to the operating day length H when the boat is used, and forces it to zero otherwise.

Constraints (10) ensure that the sum of route durations placed in shift s of boat k fits within the maximum shift length S . This per-shift constraint is what guarantees feasibility at the crew level, since routes are indivisible and cannot be split across shifts.

Constraints (11) prevent a shift from being used unless its boat is also marked as used, keeping the boat count $\sum_k u_k$ consistent with the shift assignment.

Constraints (12) and (13) break symmetry by forcing boats and the shifts within each boat to be used in increasing index order. Without them, the solver would explore equivalent permutations of the same assignment, which slows the search considerably.

The last three constraints, (14), (15) and (16), keep the decision variables in their respective domains.

Once the model is solved, the optimised schedule can be rebuilt from the decision variables to form a route-to-boat-and-shift assignment, with a total working time and a shift count $\sum_s z_{ks}$ for each boat k . As with the PSCLP, the model is implemented with Google OR-Tools (Perron and Furnon 2025) using its CP-SAT solver (Perron and Didier 2025). Route durations δ_r are converted from hours to integer seconds before being passed to the solver, since CP-SAT only accepts integer coefficients. A solver time limit of 2 min per run is set, which is well above what is needed as the number of routes per day stays low and an optimal solution is returned in under a second.

6 Results and Discussion

This section presents and discusses the results of the three models, in the order they are applied. It starts with the port selection results, covering the walking thresholds and the selected port locations. It then introduces the demand scenarios and the resulting demand at each port, before moving on to the boat routing and boat scheduling results, which give the routes, fleet and crew needed for each scenario. A summary of the main results closes the section.

6.1 Walking Thresholds

As discussed in the Methodology, the PSCLP was run with multiple combinations of walking time threshold ($T_{\max}$) and minimum population covered within that walking time (α). The best runs by utopia ranking for each $T_{\max}$ can be found in Figure 9, together with the best overall run. The two selected runs are ($T_{\max} = 15, \alpha = 0.95$) and ($T_{\max} = 28, \alpha = 0.98$), as the first is the best for $T_{\max} = 15$ and the second is the best overall run.

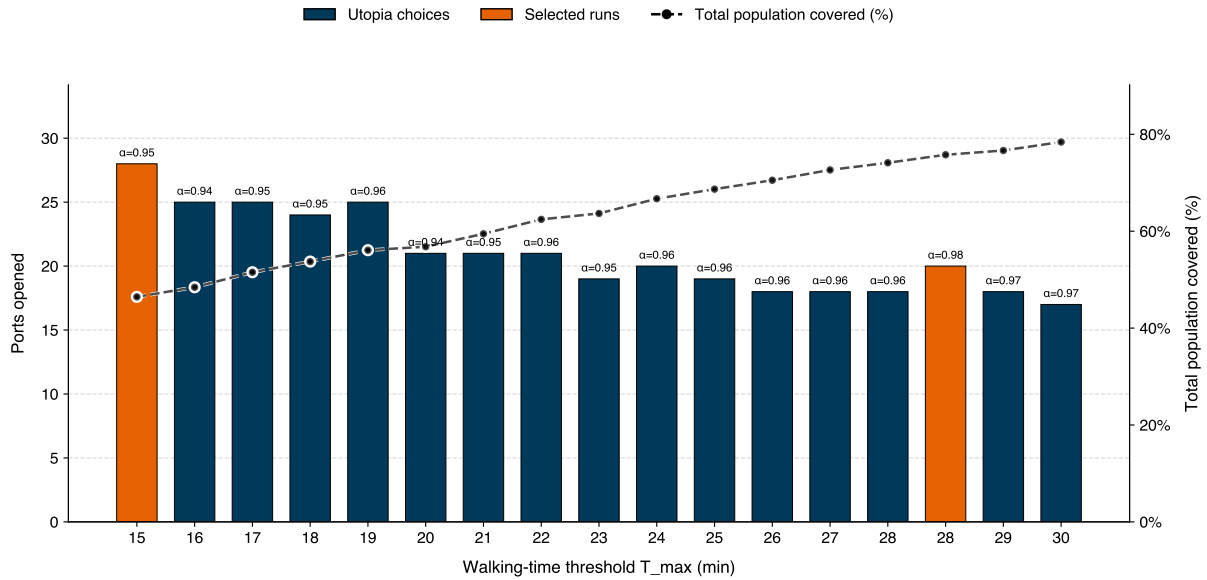


Figure 9: Summary of PSCLP utopia runs per walking threshold $T_{\max}$, with the overall utopia run.

The figure also details the relation between the number of ports needed and the walking time threshold. The smallest threshold forces the model to open many ports, but this number decreases rapidly as the threshold is relaxed, falling from 28 to 17 ports.

It can also be observed that the total population covered by $T_{\max}$ stays low for the majority

of thresholds. These results indicate that a large share of the population remains far from a candidate port by walking distance in all runs. At $T_{\max} = 15$, only 46 % of the population is within this threshold. The next 4 min add 10 %, reaching 56 % only at $T_{\max} = 19$. Relaxing the threshold to $T_{\max} = 28$ reduces the required port count to 20. This is why the utopia point method was used to select one representative run. The scenario choice involves several dimensions, which makes it difficult to identify a good compromise manually. In addition, without a financial value assigned to each opened port, or evidence on how far residents in Bergen would be willing to walk to collect a parcel, the selection has to rely mainly on quantitative run metrics.

It is worth recalling that these coverage percentages describe walking access only. As explained in Section 5.2.1, every household is still assigned to its nearest selected port, so the network carries the full demand of the included districts even when a household sits well outside $T_{\max}$.

Furthermore, the average population walking time to the nearest selected port is shown in Table A.18 in the Appendix. It stays within a small range between 19.68 and 21.92 min across the selected runs, with a standard deviation below one minute, even though $T_{\max}$ doubles from 15 to 30 min and the number of ports drops from 28 to 17. Reducing the number of ports by 11 therefore costs the average citizen around two minutes of additional walking. This suggests that the floor of the average walking time is set by Bergen's geography rather than by the strictness of the model. Further analysis could determine whether new port locations could be added or constructed to help reduce this metric.

To summarise, these runs indicate that the number of ports can be lowered at a relatively low cost, while the average walking time cannot. Since Bergen's geography sets a floor of around 20 min that none of the thresholds improve on, relaxing $T_{\max}$ and removing 11 ports only adds around two minutes to the average walking time. The choice between a denser and a sparser network is therefore mainly a question of how many ports to build, rather than how far residents have to walk.

For the rest of the thesis, the two runs are referred to as **t15** for the PSCLP configuration ($T_{\max} = 15, \alpha = 0.95$) and **t28** for ($T_{\max} = 28, \alpha = 0.98$).

6.2 Port Locations

The layout of the chosen ports for both selected runs can be found in Figure 10. It shows that the two runs have many ports in common, and both share the same general port layout, with six clearly defined zones. This indicates that some candidate ports have strong

geographic positions and can serve a large share of the population. The ports selected in $\mathbf{t28}$ are almost a subset of those selected in $\mathbf{t15}$, with only a few exceptions. This suggests that the same project could be adapted to different $T_{\max}$ values, moving from a denser port network to a more concentrated one without changing the overall modelling framework. Moreover, because the ports form clear geographic zones, some zones could be removed at the beginning and added later.

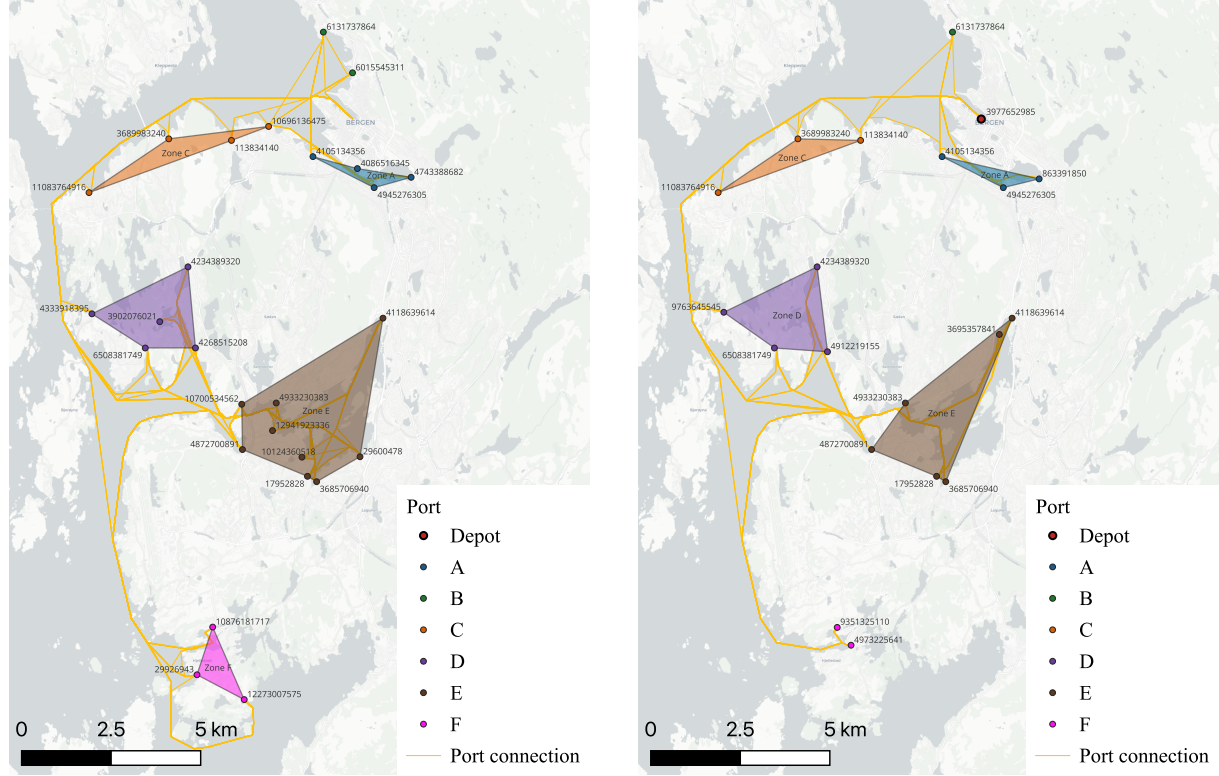


Figure 10: Selected ports and depot for PSCLP runs $\mathbf{t15}$ and $\mathbf{t28}$.

The routes drawn to calculate the distances can also be seen in Figure 10 as the orange lines on the water. Again, these lines were only used to obtain reliable distances between the ports. A real network would need to ensure that they are actually optimal routes between ports, using advanced waterway knowledge.

To better understand the effect of adding ports on population coverage, Figure 11 can be observed. It shows the shift in distribution for both runs. A large share of the population around 13 to 27 min is shifted closer to the 10 min threshold. Overall, however, both runs have similar average population walking times of 19.68 min for $\mathbf{t15}$ and 20.73 min for $\mathbf{t28}$. The cumulative population coverage by walking time, broken down by district for both runs, is shown in Figure A.22 in the Appendix.

To summarise, the selected port layout stays stable across the two thresholds. Both runs share the same six zones, and $\mathbf{t28}$ is almost a subset of $\mathbf{t15}$, which means that moving from a denser to a sparser network mostly removes ports instead of relocating them. This

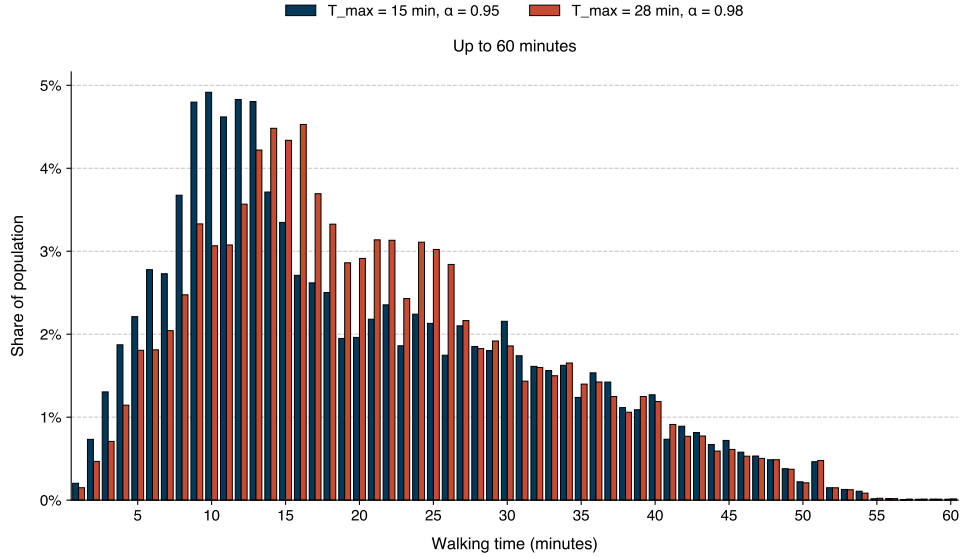


Figure 11: Total population walking time distribution for t_{15} and t_{28} .

suggests that the network could be built in stages, starting from the busiest zones and adding the others later, without changing the overall layout.

6.3 Demand Scenarios

As discussed, three demand scenarios were chosen to test the waterway network based on possible daily demands. The daily demand over the observed period and the three scenarios are shown in Figure 12, where each day is coloured according to the scenario that could handle its demand.

Two patterns stand out. First, a within-week variation is visible, with demand peaking on Tuesday and then tapering off over the rest of the week to its lowest point on Friday. This pattern is not observed in the raw data but comes from the daily demand shares introduced in Section 4.6, which spread the weekly total unevenly across the five weekdays. Second, a clear seasonality can be seen across the year. Demand rises sharply in November, most likely driven by the run-up to the Christmas period in Norway, while noticeable drops appear in April and May, and again in late July. January stands out the most, with very low demand compared to the rest of the year. This could be the result of missing data in the PostNord dataset, although the weekdays of January are indeed present for the entire month. Another explanation could be the post-Christmas slowdown, when consumer and retail activity tends to drop at the start of the new year.

The low-demand scenario corresponds to a daily network demand of 10 732 parcels. The peak scenario is more than four times higher at 44 405 parcels, while the average scenario

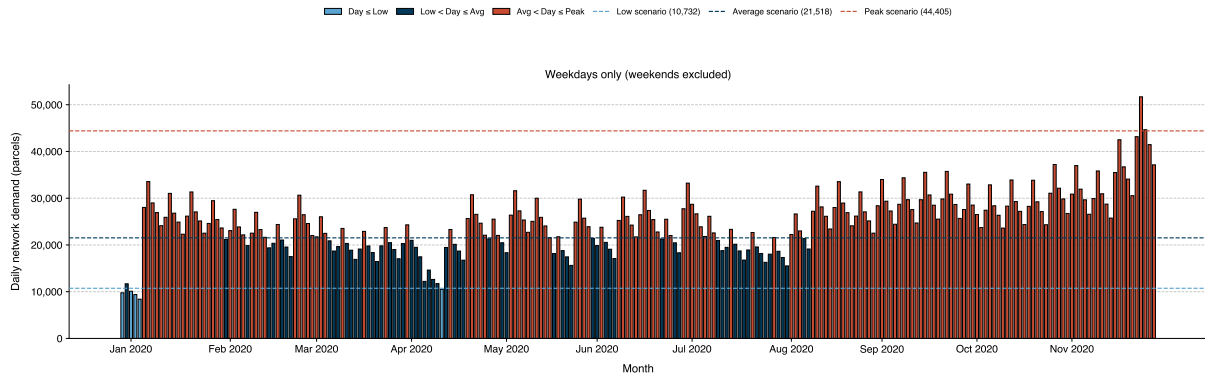


Figure 12: Daily network parcel demand over the observed period, with weekdays only, classified according to the low, average, and peak demand scenarios.

sits in between with 21 518 parcels. As shown in Figure 12, very few days in the observed year fall within the low-demand range, with most of them concentrated at the very beginning of January. The low scenario therefore serves more as a reference for the quiet end of the demand range than as a typical operating day. By contrast, the average and peak scenarios bound the majority of the actual year, with most weekdays falling between the two thresholds.

To summarise, the demand the network has to serve varies a great deal rather than staying close to a single level. The peak scenario is more than four times the low, and demand moves between the two through both a within-week variation and a clear seasonality across the year. Since the average and peak scenarios bound almost every weekday, these are the two that matter most when sizing the network, whereas the low scenario mostly marks the quiet end of the range.

6.4 Port Demands

The full demand data for each scenario and all ports can be found in the Appendix in Table A.16 for $\mathbf{t15}$ and Table A.17 for $\mathbf{t28}$. The distribution can also be visualised in Figure A.20. As discussed earlier, the depot itself also serves as a demand port for households, allowing them to pick up their parcels there. Therefore, its demand does not have to be transported by boat. In Figure 13, the depot (OSM node 3977652985), in both scenarios, has the highest percentage of total demand, followed by the same two ports. Furthermore, around 60 % of the total demand is handled by only seven ports in $\mathbf{t15}$, and 70 % in $\mathbf{t28}$. This shows that some ports are visited much more frequently than others, and that low-demand ports could potentially be removed from the network with only a limited effect on overall performance. The same demand aggregated by the geographic zones from Section 5.2.2 is shown in Figure A.21 in the Appendix.

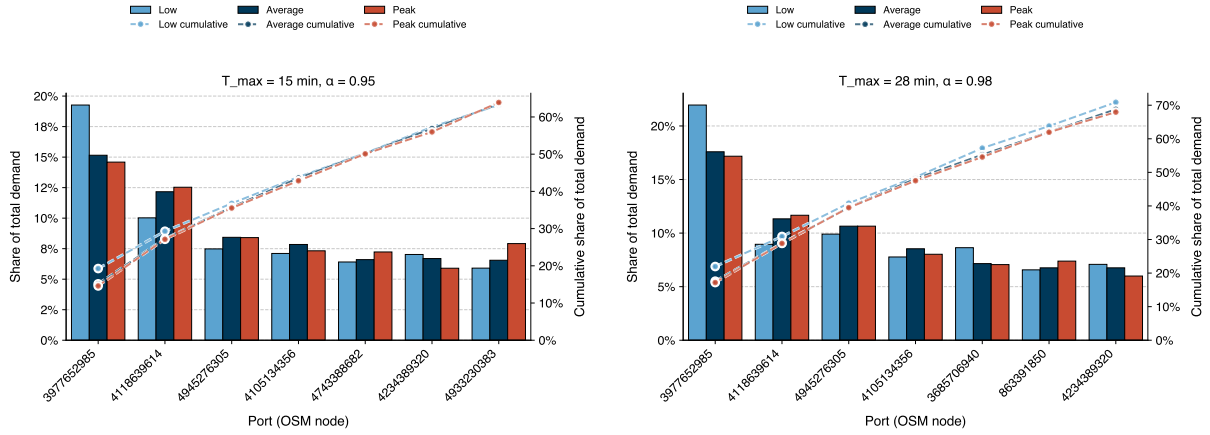


Figure 13: Distribution and cumulative share of daily parcel demand across the top seven selected ports under the low, average, and peak demand scenarios. The left graph corresponds to $T_{\max} = 15$ min and $\alpha = 0.95$, while the right graph corresponds to $T_{\max} = 28$ min and $\alpha = 0.98$.

To summarise, demand on the network is highly concentrated, with the depot and a few ports handling most of the parcels and the remaining ports each handling very little. This concentration is useful for the design of the network, since it means that the network could be introduced gradually, starting with the busiest ports. It also suggests that adapting the boat capacity to the demand of each port, rather than using a single capacity for the whole network, could improve the results.

6.5 Boat Routing

The SCVRP model returns many results to discuss. These are grouped by the selected PSCLP solution they use (**t15** or **t28**) and by configuration (see Table 6). Each selected solution produces 27 results. All configurations were run for 1 hour to give the solver sufficient time to improve the solution. Each run produced a feasible solution, although none were proven optimal by the solver. Most runs found their best solution in a matter of seconds, after which the remaining run time did not improve the solution. Others, such as **t15** with $(Q = 2500, e = 4)$ and $(Q = 1500, e = 3)$, found better solutions after around 50 min. The configurations that benefited from the full 1 h run time differed between **t15** and **t28**. In **t28**, the configurations with $Q = 500$ were the last to find their best solutions. This suggests that the simpler SCVRP formulation may have made some instances harder to improve, and that a full SDVRP formulation could be worth testing in future work.

6.5.1 Port Distances

As explained in Port Distance Matrix, some port-to-port distances were not measured directly, but were estimated through intermediate graph paths. This approximation remains acceptable for the routing analysis, not because the estimated distances are assumed to be exact, but because the optimised solutions relied only marginally on them. Across all tested scenarios, more than 95% of the distances actually used by the models corresponded to real manually measured polylines. The results are therefore mainly driven by directly measured maritime distances, while the graph-based estimates only serve to complete the matrix when no direct measurement was available.

6.5.2 Total Number of Routes

Figure 14 shows the number of routes needed for each configuration, together with the median capacity utilisation across all routes.

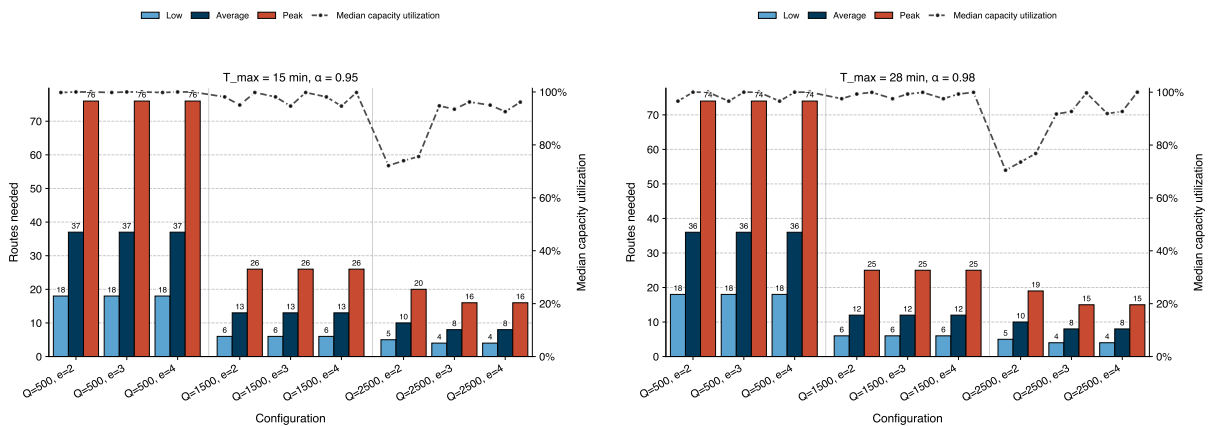


Figure 14: Number of routes required under the low, average, and peak demand scenarios, and corresponding median capacity utilisation, for each configuration and demand scenario.

The first observation is straightforward: the number of routes is inversely correlated with boat parcel capacity. As expected, the less a boat can carry on a given route, the more routes it has to run to serve the demand.

The second observation is that the number of routes is not strongly affected by the number of employees (e) in either t_{15} or t_{28} . Only $Q = 2500$ makes a difference, with both $e = \{3, 4\}$ requiring fewer routes than $e = 2$. This may seem counterintuitive, as more employees mean faster parcel handling, which should result in shorter routes and potentially different best solutions. However, the solver's objective is not to minimise the number of routes, but to minimise the total travel time for a given day. As a result, parcel handling time is not taken into account by the objective function, which explains why the

number of routes is mostly unaffected in these results. Only with the largest boat capacity, $Q = 2500$, does the reduction in handling time make enough of a difference for the solver to find a different routing network.

The third observation is that median capacity utilisation varies considerably depending on boat capacity (Q). For $Q = 2500$, in both $\mathbf{t15}$ and $\mathbf{t28}$, median utilisation is lower than for the other capacities. The lowest median, at $e = 2$, suggests that with only two employees, demand has to be split into more routes because handling time is too large. Route capacity is therefore limited not by the boat capacity, but by the speed of parcel handling. Adding one more employee ($e = 3$) greatly improves the median, bringing it above 90 % for all demand scenarios. However, the improvement stops at $e = 3$. With $e = 4$, the median remains the same as with $e = 3$ within each demand scenario for $\mathbf{t28}$ and stays around 95 % for $\mathbf{t15}$. This may indicate that $Q = 2500$ is too large for this context, as adding more employees has little impact on median capacity utilisation. The exact opposite can be seen with $Q = 500$, where median capacity utilisation sits around 95 to 100 % in all configurations and scenarios, signalling that capacity is a clear constraint because the routes are often almost full. Although a 100 % capacity utilisation rate may appear optimal in theory, it reduces operational flexibility and limits the system's ability to respond to variations in demand.

To summarise, the two extreme capacities are both poorly suited to the demand, but for opposite reasons. At $Q = 2500$, the boats are rarely filled, so the routes are limited by the 8 h duration constraint rather than by the boat capacity, which is why adding employees has little effect on the median utilisation. At $Q = 500$, the situation is reversed, as utilisation sits close to 100 % in almost every configuration and many more routes are needed to serve the same demand, which indicates that the capacity is too small. These conclusions should, however, be read with some caution for $Q = 2500$, since it produces very few routes, whereas the larger number of routes at $Q = 500$ makes its high utilisation more reliable.

6.5.3 Total Route Duration

The next metric is the total travel time of the routes. In Figure 15, these durations are shown by run and configuration, with a visual split between total navigation time and parcel handling time. Lines representing the median route time in hours are also present for each demand scenario.

The first observation is clear: as with the total number of routes needed, total route time increases across all configurations as demand scenarios rise. This observation, paired with

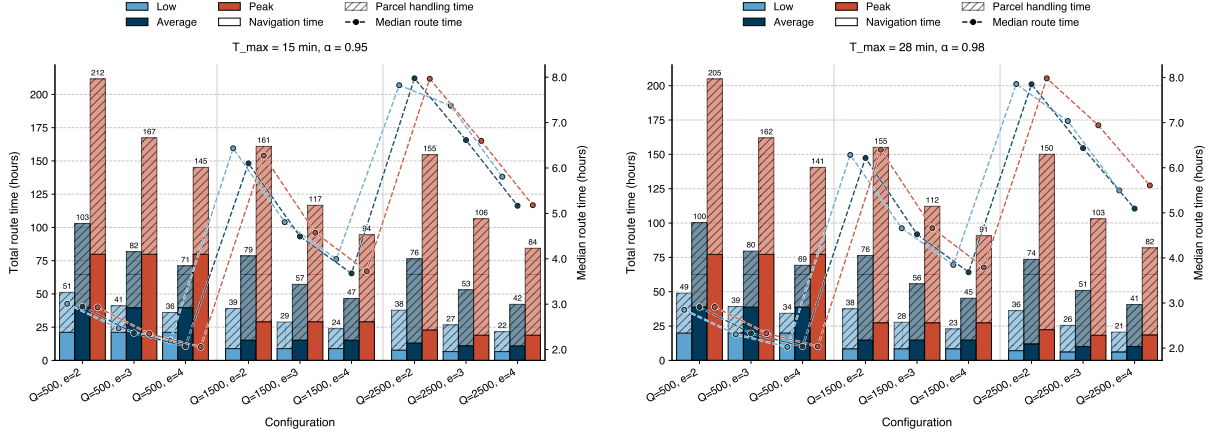


Figure 15: Total route time by configuration and demand scenario, separated into navigation time and parcel handling time.

the seasonality seen in Figure 12, shows how much the system would have to expand to handle peak demand. However, in these periods, the number of employees per boat could be increased to help reduce overall route time.

The second observation is that crew size affects total route duration more clearly than it affects the number of routes. For example, in τ_{15} , all crew sizes produced the same number of routes per demand scenario. Even with the same route count, total route duration is lower with larger crews.

A third observation is that the reduction in total route time is not directly proportional to the number of employees added to the crew. For example, in τ_{28} , going from $e = 2$ to $e = 3$ reduced the average total route time by 26 %, while going from $e = 3$ to $e = 4$ only achieved a reduction of 17 %. This diminishing return follows directly from the model, since only the parcel handling portion of the total time scales with e . The handling time at a port is $d_i\tau/e$, while the navigation and manoeuvring times do not depend on e at all. The same pattern also explains why the reduction is stronger at higher boat capacities, where each route carries more parcels and the handling share of the total time is larger, leaving more work for the additional crew member to absorb.

A fourth observation is that the reduction in total route time from increasing the boat capacity Q is also not linear. Going from $Q = 500$ to $Q = 1500$ reduced the average total route time by 29 %, while going from $Q = 1500$ to $Q = 2500$ only achieved a reduction of 7 %. This is consistent with the capacity utilisation results in Figure 14, which already showed that the largest boats were not filled to their full capacity. The extra space offered by $Q = 2500$ over $Q = 1500$ is therefore mostly unused and brings little additional benefit for total route time.

A final observation can be made about median route duration. Looking at $Q = 500$ reveals that the routes are very short, only 2 to 3 h, while the maximum constraint is set at 8 h. This shows that with this capacity, the routes are limited not by travel time but by boat capacity, which further supports the capacity utilisation results in Figure 14. The opposite can be observed for $Q = 2500$.

To summarise, the total route time follows a clear pattern of diminishing returns for both the crew size e and the boat capacity Q . Parcel handling dominates the total time and is the only component that responds to a larger crew, while the navigation time stays fixed regardless of the configuration. Each new employee, or each step up in capacity, therefore contributes less than the previous one, and the largest boats remain partly empty. Combined with the strong seasonal variation in demand, this suggests that the most efficient strategy is a balance rather than a maximum, with the crew size potentially adjusted between low and peak periods instead of permanently oversizing the fleet.

6.5.4 Total Route Distances

The next metric is total route distance in nautical miles. This metric matters because, in a real scenario, boats have a maximum range and need to be refuelled or recharged. However, as previously stated, the model does not include any constraints on maximum route distance, only on duration and total capacity. Figure 16 shows the value for this metric.

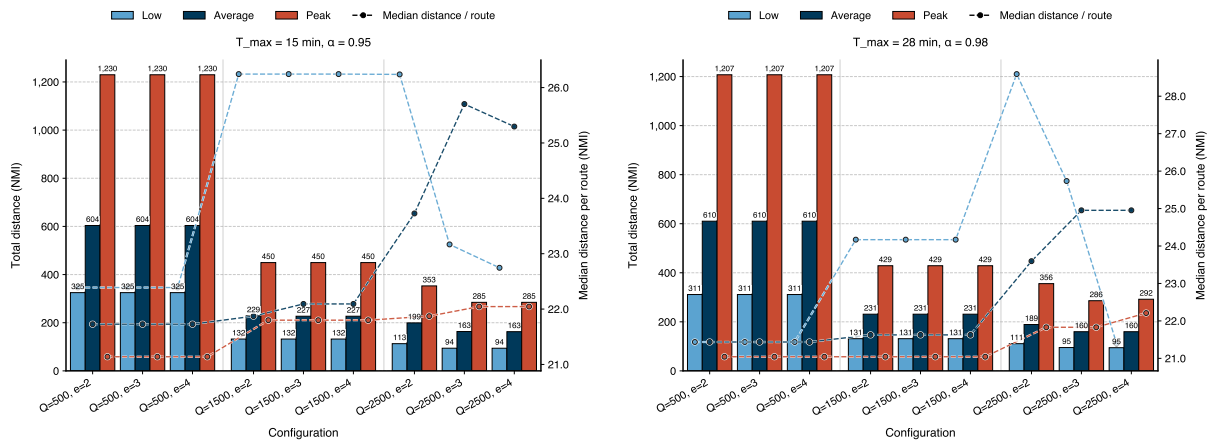


Figure 16: Total sailing distance and median distance per route for each configuration and demand scenario.

The first observation is that total distance drops drastically between $Q = 500$ and $Q = 1500$. This is explained by the decrease in the total number of routes, as seen in Figure 14. More routes for the same total demand mean more navigation to and from the depot,

which serves no demand and rapidly increases distance. Having more capacity therefore reduces the number of routes, which tends to reduce total distance travelled. In contrast, as with the other metrics discussed previously, the changes from $e = 2$ to $e = 3$ are much larger than those from $e = 3$ to $e = 4$. This again points to $Q = 500$ being too small and $Q = 2500$ too large.

The second observation is that the median distance per route behaves differently across scenarios. For average and peak demand, it stays between 21 to 22 nmi for $Q \in \{500, 1500\}$ in both runs. At $Q = 2500$, the average-demand scenario climbs quickly, while the peak-demand scenario keeps a slow and steady rise. In contrast, the low-demand scenario is less predictable. At $Q = 1500$ its median stays flat across all values of e , around 26 nmi in **t15** and 24 nmi in **t28**. This suggests that the typical route distance is relatively stable regardless of e . At $Q = 2500$, **t15** and **t28** diverge. For low demand, **t15** drops to about 23 nmi, while **t28** climbs to a peak of more than 28 nmi at $e = 2$ before falling back to around 26 nmi. The average scenario also splits. In **t15** it peaks at 25.7 nmi at $e = 3$ and drops slightly at $e = 4$, while in **t28** it peaks at a much lower 25 nmi for $e = 3$ and stays mostly unchanged for $e = 4$. Most of this variation comes from the small number of routes at $Q = 2500$. With only around 5 routes, removing a single route of 27 nmi between $e = 2$ and $e = 3$ is enough to shift the median by several nautical miles.

To summarise, total distance is the metric that separates the three capacities most clearly, and it again points to $Q = 1500$ as the most efficient choice. The distance drops drastically from $Q = 500$ to $Q = 1500$, since fewer routes mean less navigation to and from the depot, which serves no demand. Increasing the capacity further to $Q = 2500$ brings little additional benefit, as the extra space is mostly unused. The number of employees has almost no effect, since the model minimises navigation time rather than distance, so capacity remains the only real driver, and $Q = 1500$ already captures most of the possible reduction.

6.6 Boat Scheduling

The next step was to optimise the assignment of routes to boats and shifts. Unlike the port-distance step after the PSCLP, this model does not require any additional manual information, so it was run on all route solutions. The two key metrics discussed are the number of boats and the number of employees per day needed to serve the routes for each configuration, for both **t15** and **t28**. Both metrics are shown in Figure 17.

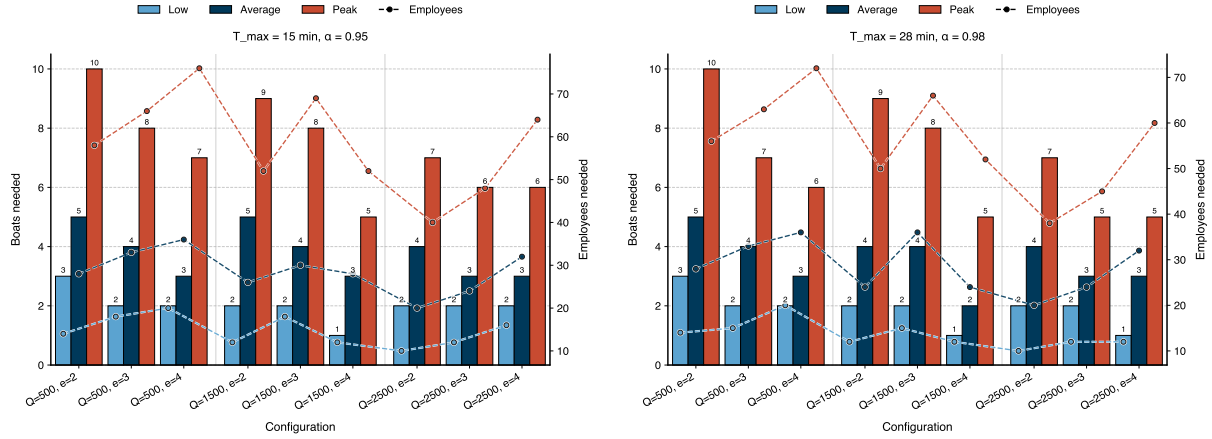


Figure 17: Number of boats and employees needed for all configurations and demand scenarios.

6.6.1 Boats Needed

The first observation concerns the low number of boats needed compared to the number of routes from Figure 14. For example, with $Q = 500$ at peak demand, all 76 routes are covered by only 10 boats. This works because the routes are very short, so many of them can be chained on the same boat within the 24 h daily window. As routes get longer, fewer of them fit on a single boat. The variation in boat count between demand scenarios stays high, although it shrinks when more employees are assigned per boat, since a larger crew shortens each route.

The second observation is that the number of boats needed across Q values within the same demand scenario stays similar for both low and average demand scenarios. Indeed, the standard deviation stays below 1, with values between 1 to 3 boats for low demand and 2 to 5 boats for average demand. The peak scenario shows more variation, with the boat count ranging from 5 to 10 and a standard deviation of about 1.8. On top of that, both t_{15} and t_{28} require a very similar number of boats, which means the capital cost of the fleet cannot really be used to decide between the two routing scenarios. Moving from average to peak demand, however, increases the boats needed by 95 % on average, which confirms that demand level is a real driver of fleet size.

The final observation for the boats needed is that the number of employees per boat still plays a role in reducing the boat count in higher-demand scenarios, while reducing it by only 1 in the low-demand scenario. This still represents around 33 to 50 %, depending on boat capacity. Nevertheless, for all scenarios, even if the smallest crew size is used for the low-demand scenario and the highest for the peak scenario, the number of boats would still have to double at least to serve all demand scenarios for a given Q .

To summarise, the number of boats needed is determined mainly by the demand scenario rather than by the other design choices. It stays very similar between **t15** and **t28**, and across boat capacities, which means that neither the port layout nor the choice of Q can be decided on fleet size alone. Demand, on the other hand, is a real driver, since the fleet has to roughly double from the average to the peak scenario. The main question for an operator is therefore how much capacity to provide for peak periods, rather than which of the two networks to operate.

6.6.2 Employees Needed

This metric helps identify the impact that different values of e have on the total employee count for the day. The first observation is that increasing e almost always increases the total number of employees needed, with one exception at $Q = 1500$. For this boat capacity, in every demand scenario, going from $e = 2$ to $e = 3$ raises the employee count, but going from $e = 3$ to $e = 4$ lowers it by an average of 22 %, while also reducing the number of boats needed. As shown in Figure 14, the total number of routes did not change with different values of e for this Q . However, Figure 15 showed that both the total and median routing time decreased. This means that the extra employees made the routes shorter, which allowed the same crew to cover more routes during the day, and therefore reduced the total employee count.

For the other values of Q , the behaviour is more straightforward. Going from $e = 2$ to $e = 3$ increases the total employees needed by an average of 20 %, while reducing the number of boats by 18 % on average for $Q = 500$ and 7 % for $Q = 2500$. For $Q = 2500$, the boat reduction only happens between $e = 2$ and $e = 3$ and disappears between $e = 3$ and $e = 4$. This shows that even when adding more employees makes the routes shorter, the gain is not always large enough to free up an entire boat.

In a real-world setting, this trade-off between fleet size and crew size could be settled on a cost basis, as long as the daily operating cost of a boat and the daily cost of an employee are both known. Without those figures, only assumptions can be made here, and the best configuration remains dependent on the cost structure a future operator works with.

6.7 Summary of Results

In summary, since both **t15** and **t28** serve the same total amount of demand, the logistics between the two scenarios do not change very much. They would require similar numbers

of boats and employees, travel roughly the same distances, take the same amount of time, and use a similar number of routes. These results can be seen in Table 9. Nevertheless, without any cost information, the average percentage changes for the variables related to financial cost (employees, route distance and route duration) are always lower or at least equal in **t28** compared to **t15**. This suggests that the operating cost of **t28** would be lower than that of **t15**. The full metric results, averaged over e , are shown in Table 10.

Table 9: Average percentage change in routing metrics when comparing **t28** to **t15**, by demand scenario and overall. Negative values indicate lower values for the 28-minute solution.

Variable	Low	Average	Peak	Overall
Routes	–	-3%	-4%	-3%
Route duration	-4%	-3%	-3%	-4%
Route distance	-2%	0%	-2%	-1%
Boats	-6%	-6%	-7%	-6%
Employees	-6%	–	-4%	-4%
Median capacity utilisation	-2%	1%	1%	–
Median route time	-4%	-1%	2%	-1%
Median distance per route	-2%	-2%	-1%	-2%

Table 10: Average routing performance metrics for the **t15** and **t28** port-location solutions, reported by demand scenario.

Variable	Low		Average		Peak	
	t15	t28	t15	t28	t15	t28
Routes	9.44	9.44	19.56	18.89	39.78	38.44
Route duration	34.00	32.57	67.74	65.76	137.96	133.39
Route distance (nmi)	185.86	180.90	335.51	336.97	662.33	649.13
Boats	2.00	1.89	3.78	3.56	7.33	6.89
Employees	14.67	13.56	28.56	28.56	58.33	55.78
Median capacity utilisation	95 %	93 %	94 %	95 %	96 %	97 %
Median route time (hours)	4.88	4.71	4.60	4.56	4.63	4.74
Median distance per route (nmi)	24.23	23.62	22.89	22.52	21.64	21.35

7 Conclusion

This final section concludes the thesis. It first returns to the research question and answers it in three parts, drawing on the results of the three models. It then outlines the main limitations of the work and the directions that future research could take.

7.1 Answer to the Research Question

With the results established, the research question asked at the beginning can now be answered in three parts.

How can a waterborne parcel delivery network be designed for the city of Bergen, and what fleet and crew would such a network need to operate?

The final model does not include every district in Bergen. Arna, Fana, and Åsane were excluded after the walking-time analysis showed that the candidate ports could not provide realistic access in those districts. The results should therefore be interpreted as a network for the central and coastal part of Bergen, rather than for the whole municipality.

1. How many ports are needed, and where should they be placed, to cover the modelled population within a reasonable walking time?

Two reasonable walking times were chosen, 15 and 28 minutes. For the 15-minute threshold, 28 ports were needed to cover 95 % of the population reachable within that walking time. For the 28-minute threshold, 20 ports covered 98 % of the population reachable within that walking time. The first scenario demonstrates a strict walking-time target with a denser port network, while the latter serves as a compromise between the number of ports and the walking distance. However, the results indicate that both port scenarios lead to similar average walking times to the nearest port. The 15 min scenario has an average walking time of 19.68 min, compared with 20.73 min for the 28 min scenario. This suggests that, in practice, the difference between the two networks is smaller than the $T_{\max}$ values alone might suggest.

2. Given a set of selected ports and their parcel demand, what set of boat routes can serve them while respecting realistic capacity and shift limits?

Three demand scenarios were tested, representing low, average and peak demand periods. For both port scenarios, the number of routes was similar; the real driver was demand. The number of routes ranges from only 4 to 18 for low demand depending on capacity, 8 to 37 for average demand and 15 to 76 for peak demand.

3. Given these routes, how many boats and crews are required to operate the network across low, average and peak demand days?

For both walking distance scenarios, the number of boats and employees is extremely similar, varying again mostly based on the demand scenarios. For low demand, the fleet ranges from 1 to 3 boats, with 10 to 20 employees. For average demand, the fleet ranges from 2 to 5 boats, with 20 to 36 employees. For peak demand, the fleet would range from 5 to 10 boats, with 38 to 76 employees.

In conclusion, t_{15} and t_{28} ended within a few percent of each other on every metric, which means the two layouts would be similar to operate. The results, however, show the extent to which the operations would have to scale to accommodate low and peak demand. Furthermore, the results show that a boat capacity of 1,500 parcels would be more efficient than the low and high capacities, offering more flexibility than the 500-parcel capacity with better overall capacity utilisation. For crew size, three employees helped reduce the number of boats needed, while increasing the crew to four improved peak-demand performance.

7.2 Further Research

Although the scenarios in this thesis were developed from available data, literature, and realistic operational assumptions, they remain simplified representations of a larger implementation problem. The results should therefore be read as a first estimate of the requirements for a waterborne parcel delivery network in Bergen, rather than as a final operational design. Several aspects can still be refined or tested further.

The most direct extension is to add cost data. With a daily cost per boat, per employee, and a fixed cost per port, the choice between t_{15} , t_{28} , and the (Q, e) configurations becomes a simple optimization on top of the existing results.

A second direction is to refine the walking time by accounting for elevation. Bergen is hilly, and a slope-corrected travel time over the full round trip to the port would likely change which ports the PSCLP selects, especially in the steeper districts.

A third direction is to extend the access measure beyond walking. This thesis assumes that residents reach their port on foot, which becomes unrealistic for households far from any port. Adding other modes, such as cycling or public transport, would give a more complete picture of how residents actually reach their port and of the true access effort across the network, especially in the areas that walking alone covers poorly.

A fourth direction would be to extend the model to a heterogeneous fleet setting, where boats with different parcel capacities can be used in the same solution. This could improve the routing results, since larger boats may be useful for high-demand ports, while smaller boats may be better suited to lower-demand ports. A combination of capacities could therefore create a more efficient network by matching boat size more closely to port demand.

A fifth direction is to validate the port candidates on the ground, since some of them may be private, too shallow, or otherwise unsuitable in ways that open data cannot detect.

Finally, the environmental side of the project, which motivated this thesis but was kept outside its analytical scope, can now be quantified using the route distances, durations and fleet sizes produced here, and compared against the emissions of the boats themselves.

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Declaration on the use of AI tools in the work on this master's thesis

Name (and version) of the AI tool: Gemini 3.1 Pro

Purpose of using the tool: Used to assist with LaTeX formatting.

Name (and version) of the AI tool: ChatGPT GPT-5.5 and Claude Opus 4.7

Purpose of using the tool: Used across the thesis to assist with grammar, syntax, source exploration, and document structure. These models were also used for support while developing the code necessary for this research.

I am aware that I am responsible for all content of this master's thesis, including the parts where AI tools are used. I am responsible for ensuring that the thesis complies with ethical rules for privacy and publication.

A Appendix

A.1 Population

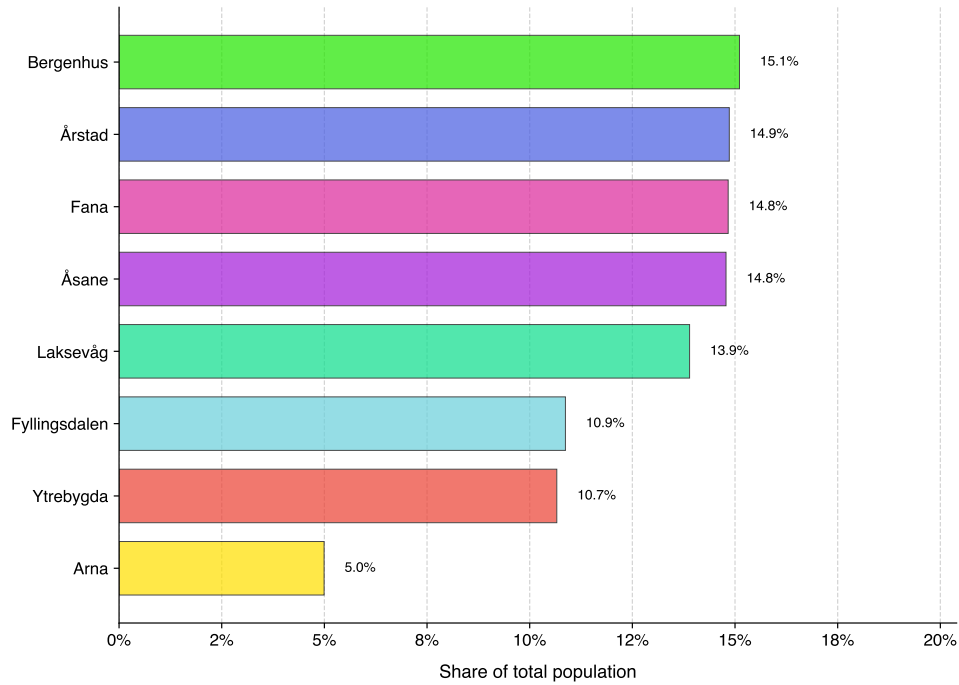


Figure A.18: City of Bergen population distribution by district in 2020.

A.2 Ports

Table A.11: OpenStreetMap tags used to extract candidate ports.

Key	Value	Description
amenity	ferry_terminal	Ferry terminal/stop.
man_made	pier	Raised walkway over water.
man_made	quay	Structure alongside navigable water used for loading vessels.

A.3 Demand

$$\text{CAGR} = \left(\frac{D_{2025}}{D_{2022}} \right)^{\frac{1}{3}} - 1 \approx 4.68\% \quad (\text{A.17})$$

A.4 Results

Table A.12: Total OSM candidate port nodes aggregated by tag.

Infrastructure	Node Count
pier	286
ferry_terminal	10
quay	10
mixed*	4

* A node is classified as mixed when port locations of different tags were snapped to the same OSM node.

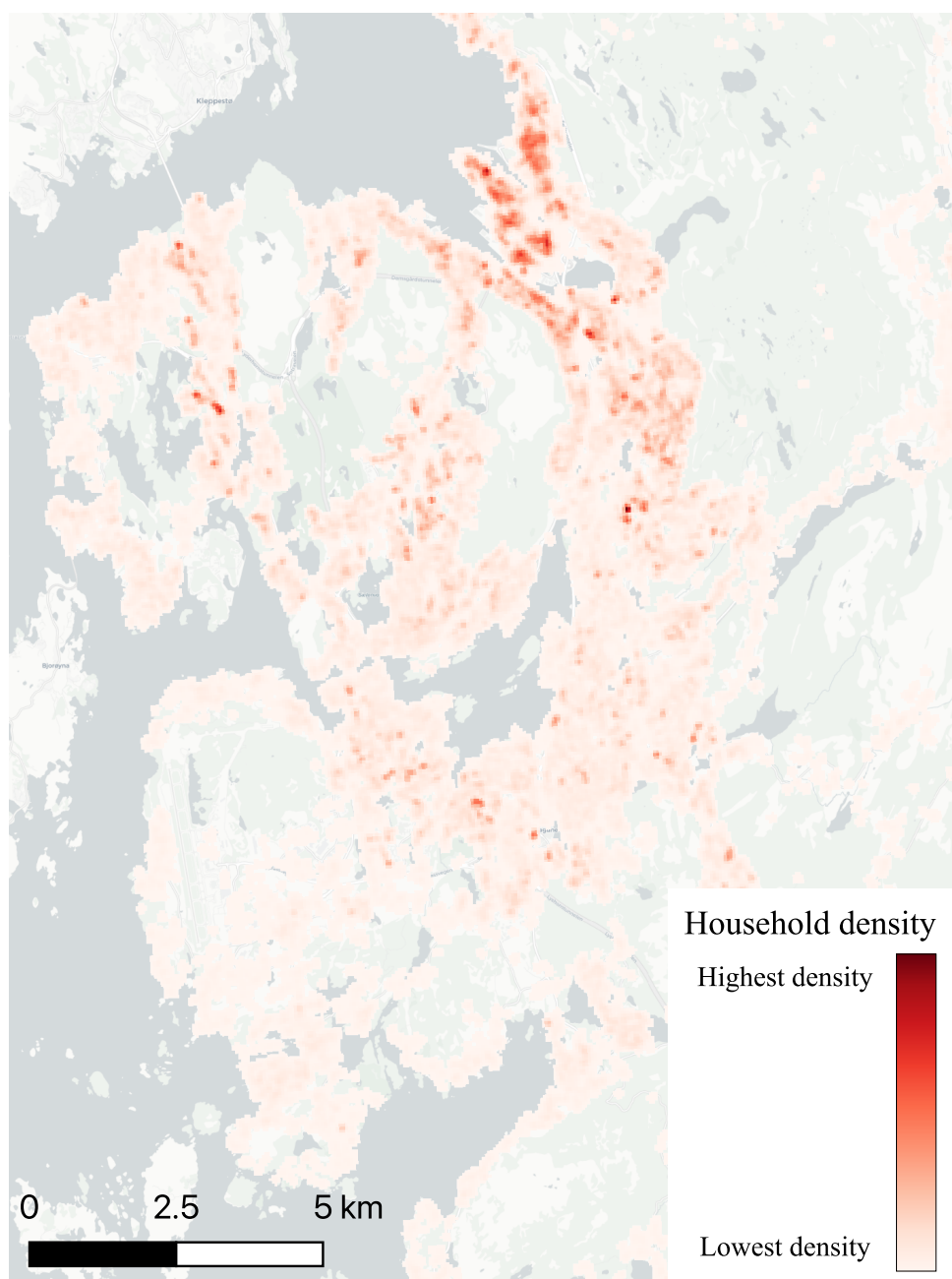
**Figure A.19:** City of Bergen population density in 2020.

Table A.13: Private market consumer parcel courier usage (Nkom 2026).

Company	Survey Value (%)	Normalised Value (%)
Posten Norge AS/Bring AS	85	55.56
PostNord AS	30	19.61
Helthjem AS	21	13.73
Vet ikke	8	5.23
Instabox Norway	4	2.61
Andre	3	1.96
Porterbuddy	2	1.31

Table A.14: PostNord annual parcel demand (PostNord AB 2021, 2022, 2023, 2024, 2025, 2026).

Year	Parcel Demand (million)
2020	198
2021	226
2022	245
2023	243
2024	245
2025	281

Table A.15: 7-day weekly demand split (Lorenzo-Espejo et al. 2024) and new 5-day split.

Day of Week	7-Day Split	5-Day Split*
Monday	0.1965	0.1979
Tuesday	0.2370	0.2370
Wednesday	0.2047	0.2047
Thursday	0.1901	0.1901
Friday	0.1703	0.1703
Saturday	0.0014	—
Sunday	0.0000	—

* The 7-day split's Saturday volume is given to Monday's allocation for the 5-day weekday operations.

Table A.16: Port demand scenario data for run $T_{\max} = 15$ and $\alpha = 0.95$ (in parcels).

OSM Node	Low Scenario	Avg Scenario	Peak Scenario
17952828	96	167	349
29600478	232	366	720
29926943	50	95	248
113834140	18	59	131
3685706940	696	1,168	2,417
3689983240	40	64	189
3902076021	16	39	117
3977652985	2,068	3,259	6,486
4086516345	421	768	1,598
4105134356	763	1,687	3,256
4118639614	1,076	2,616	5,571
4234389320	754	1,439	2,622
4268515208	101	170	409
4333918395	115	248	565
4743388682	688	1,420	3,214
4872700891	499	1,191	2,271
4933230383	634	1,409	3,519
4945276305	803	1,813	3,737
6015545311	274	622	1,267
6131737864	392	725	1,356
6508381749	94	203	480
10124360518	11	23	60
10696136475	75	191	360
10700534562	128	223	522
10876181717	296	710	1,253
11083764916	322	681	1,397
12273007575	21	40	106
12941923336	51	111	233

Table A.17: Port demand scenario data for run $T_{\max} = 28$ and $\alpha = 0.98$ (in parcels).

OSM Node	Low Scenario	Avg Scenario	Peak Scenario
17952828	104	187	400
113834140	41	134	285
863391850	705	1,454	3,279
3685706940	927	1,539	3,135
3689983240	40	64	189
3695357841	125	194	445
3977652985	2,356	3,785	7,627
4105134356	834	1,837	3,563
4118639614	961	2,438	5,180
4234389320	760	1,454	2,661
4872700891	600	1,429	2,739
4912219155	240	424	951
4933230383	607	1,351	3,410
4945276305	1,063	2,291	4,728
4973225641	124	219	523
6131737864	506	1,054	1,918
6508381749	103	228	540
9351325110	199	507	871
9763645545	118	255	578
11083764916	319	674	1,383

Table A.18: PSCLP utopia choices by walking-time threshold.

$T_{\max}$	α	Ports Needed	Population Covered	Avg Walking Time (min)
15	0.95	28	46.45%	19.68
16	0.94	25	48.43%	19.97
17	0.95	25	51.53%	19.92
18	0.95	24	53.73%	20.09
19	0.96	25	56.08%	19.91
20	0.94	21	56.80%	20.29
21	0.95	21	59.46%	20.34
22	0.96	21	62.44%	20.51
23	0.95	19	63.67%	20.88
24	0.96	20	66.70%	20.71
25	0.96	19	68.68%	20.83
26	0.96	18	70.52%	21.09
27	0.96	18	72.65%	21.09
28	0.96	18	74.13%	21.19
28	0.98	20	75.78%	20.73
29	0.97	18	76.66%	21.44
30	0.97	17	78.42%	21.92

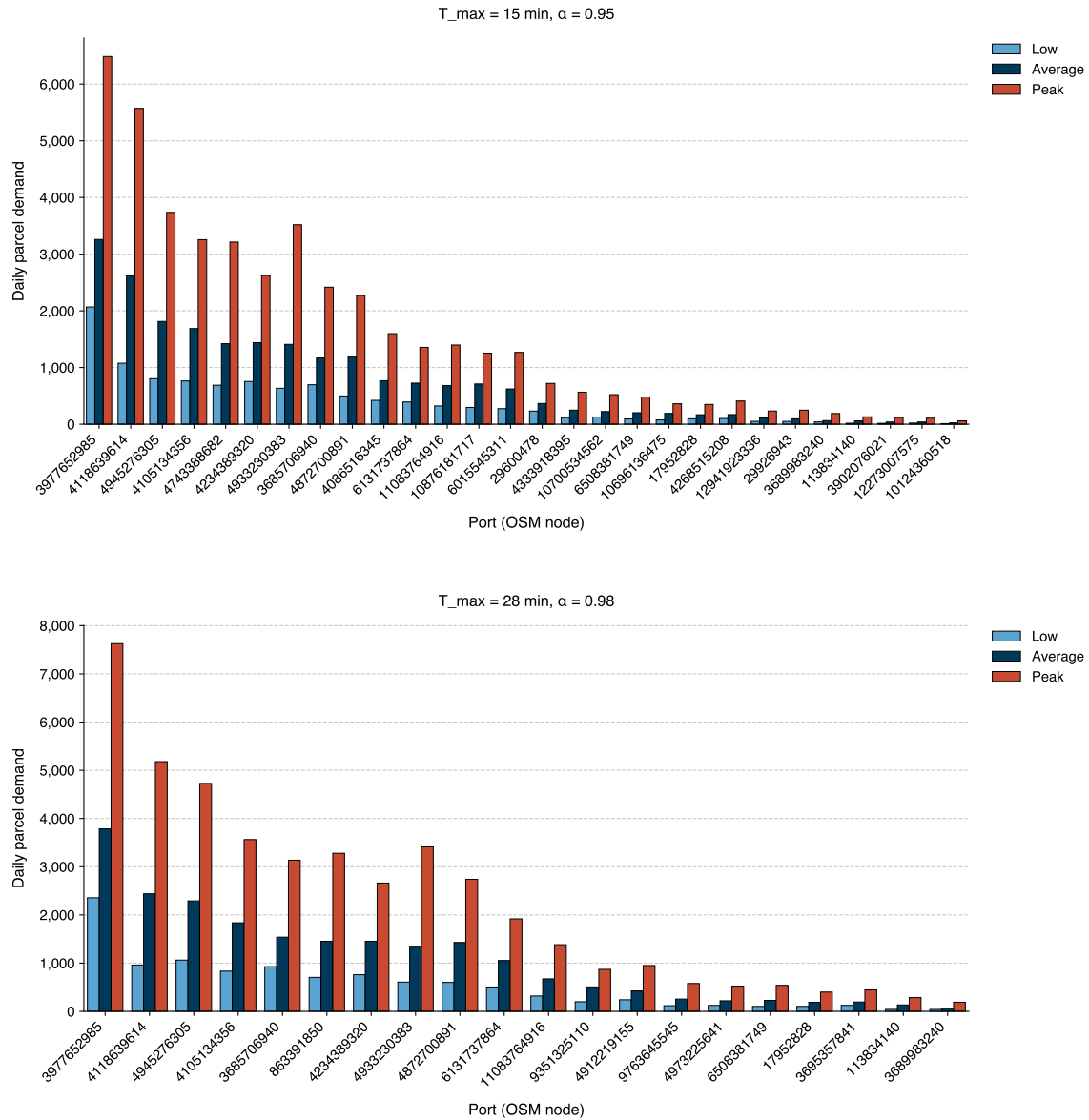


Figure A.20: Distribution of daily parcel demand across selected ports and the depot under the low, average, and peak demand scenarios. The upper graph corresponds to $T_{\max} = 15 \text{ min}$ and $\alpha = 0.95$, while the lower graph corresponds to $T_{\max} = 28 \text{ min}$ and $\alpha = 0.98$.

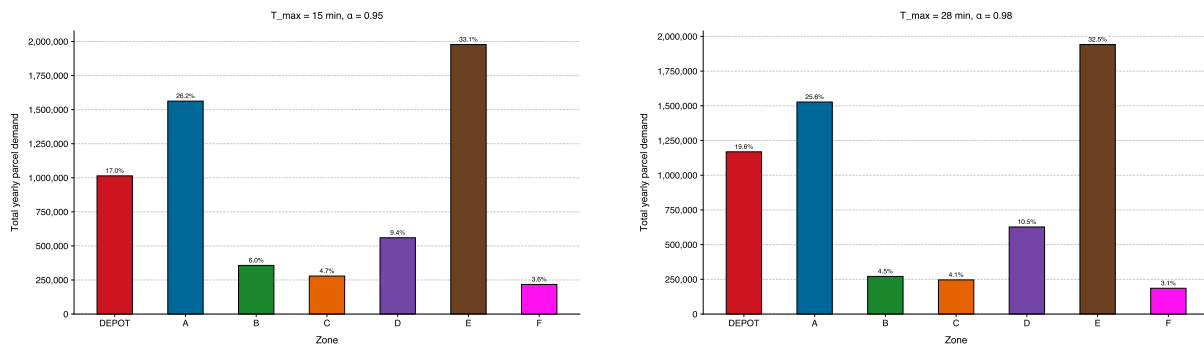


Figure A.21: Yearly demand by port zones.

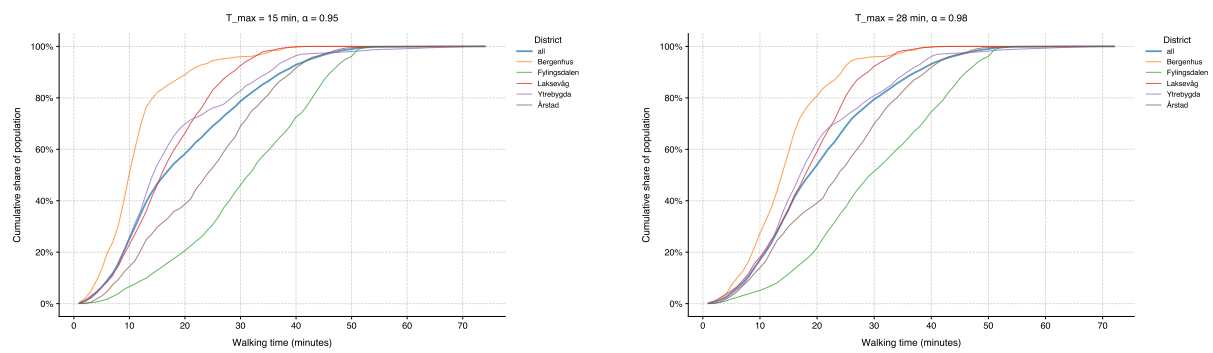


Figure A.22: Cumulative total population coverage, grouped by district and overall, by walking time.